AE 451 IRRIGATION AND DRAINAGE

A course note prepared for agricultural engineering students of the Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

Abstract

This course note presents principles of irrigation and drainage engineering with a systematic approach to designing such systems. It is presented in simple language and recommended to engineering students and practitioners

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Module 1: Introduction

1.1 Irrigation: Definition and Importance

Definition: Irrigation is the artificial application of water to the soil for crop production. Naturally, the soil medium is used for plant production but soil-less media and other media can also be used. Irrigation is not a substitute for rainfall but it is intended to supplement rainfall. But in areas where rainfall is too small or virtually non-existent, irrigation can be used to provide all the crop water requirements.

Importance of Irrigation: Some important reasons why irrigation should be undertaken are as follows:

a) About twenty percent (20 %) of the world's arable land is said to be under irrigation, yet provides 40 – 50 % of the world's food and fiber requirements. Irrigation therefore can produce more crop yield per unit area of land.
b) Irrigation ensures crop and food security and improves farm income and the economy of a nation.
c) It allows for a more efficient use of water for crop production. Crop water productivity is improved, implying more crops per amount of water used.

1.2 Supplementary and Total Irrigation

Irrigation can be classified into two depending on rainwater availability: 1) supplementary irrigation; and 2) total irrigation.

Supplementary irrigation: This is practiced in the humid and semi-humid areas of the world and suggests that crop water requirements are met through precipitation and irrigation. It can be expressed mathematically as:

$$\textit{Crop water requirements} = \textit{Rainfall} + \textit{Irrigation} \qquad 1.1$$

Total irrigation: This requires providing full crop water requirements from irrigation. It is practiced in the arid and semi-arid areas where rainfall amount is too small or virtually non-existent. It can be expressed mathematically as:

$$\textit{Crop water requirements} = \textit{Irrigation} \qquad 1.2$$

1.3 Gravity and Pumped Irrigation

Irrigation schemes can be classified depending on the mode of water delivery: 1) gravity schemes; 2) pumped schemes.

Gravity schemes: For the gravity schemes, water delivery is entirely by gravity from the source to the farm and should happen only if the farm level is lower than the level of water source.
Pumped schemes: In this case, water delivery is by pumping under pressure from the source to the farm. If farm level is higher than the level of water source, the water is delivered by pumping.

Gravity and pumped schemes: On the same irrigation scheme, it is possible that water delivery can be done by both gravity and pumping depending on the landform. Those schemes are therefore referred to **as** gravity and pumped schemes.

1.4 Irrigation in Ghana

The Ghana Irrigation Development Authority (GIDA) is responsible for the development and management of irrigation in Ghana. Putting irrigation in the context of Ghana, there is the need to briefly explain the hydrology and agroecology of Ghana and the policies guiding the practice.

Hydrology of Ghana: The water year in Ghana begins on 01 March of each year and ends on 28/29 February of the following year. The water regime in the country changes according to the seasons. Officially, the rains begin at the beginning of March of each year even though this can be delayed 1 to 3 months as one moves inland from the coast. In the same vein, the dry season officially ends at the end of February of each year even though this can be delayed for 2 to 3 months as one moves up north. All water sources (surface + ground water) are at the lowest levels at the peak of the dry season in Ghana.

Ghana is drained by 3 drainage systems: 1) the Volta river system; 2) the south-western river system; and 3) the coastal river system (Table 1.1).

Table 1.1 The Drainage System of Ghana

River System	Constituent Rivers	% Runoff Contribution	Mean Runoff (billion m^3/y)
Volta	White Volta, Red Volta, Black Volta, Oti	65	35.4
South-western	Bia, Tano, Birim, Pra	29	15.8
Coastal	Tordzie/Aka, Densu, Ayensu, Ochi-Nakwa , Ochi-Amissah	6	3.2
Ghana		**100**	**54.4**

Ghana is underlain by three main geological formations, namely the basement complex comprising crystalline igneous and metamorphic rocks, the consolidated sedimentary formations, underlying the Volta basin and the Mesozoic and Cenozoic sedimentary rocks. These formations cover 54 %, 45 % and 1 % of the country respectively. The rocks of the basement complex and the Voltaian formation have little or no primary porosity. Ground water occurrence is associated with the development of secondary porosity brought about by shearing, fracturing and weathering. The depth of aquifers, are normally between 6 and 60 m with yields rarely exceeding 6 m^3/h.

Most irrigation systems in Ghana are derived from surface water systems either by direct abstraction or from reservoirs created by constructing dams across seasonal streams. Groundwater use for irrigation in Ghana is limited and dispersed across the country with the south-east [Keta] and north-east [Bawku] parts being the most notable.

Agroecology of Ghana: Ghana [238,535 km^2] has 6 distinct agro-ecological zones, according to the rainfall distribution in the country (Table 1.2):

Table 1.2: Agroecology of Ghana

Agroecology	Area (km^2)	Average Rainfall (mm)
Rainforest	7,156	2,200
Deciduous Forest	66,790	1,500
Transitional Zone	7,157	1,300
Coastal Savannah	4,770	800
Guinea Savannah	150,277	1,100
Sudan Savannah	2,385	1,000
Ghana	**238,535**	**1,300**

Irrigation activities are planned according to the seasons in Ghana where cropping is done mainly under rainfed conditions during the rainy season and irrigation becomes more pronounced in the dry season. Some form of irrigation is practiced in all the agroecological zones but the main irrigation activities are located in the Guinea, Sudan and coastal savannahs where substantial water deficits are experienced in the year.

Irrigation Policy in Ghana: Irrigation policy document classifies irrigation development in Ghana as: 1) formal; 2) informal; and 3) commercial.

Formal schemes: These are government developed schemes under GIDA management. Apart from those conceived, designed and developed by GIDA, there are also the bilateral and multi-lateral project-linked schemes that are developed as poverty-reduction intervention schemes. Land area under this type of irrigation ranges between 8,000 and 10,000 ha.
Informal schemes: These include private, individual or community-initiated schemes usually at the subsistence level. These are located in: 1) areas where farmers have access to shallow groundwater; 2) low-lying areas where farmers take advantage of the residual moisture in the soil for crop production; and 3) areas where farmers can pump water directly from nearby surface sources as streams and ponds for crop production. The actual area under the informal schemes is not known but could be in the range of 200,000 ha and above.
Commercial schemes: These are the private, export-led commercial irrigators. They invest in the more sophisticated, capital-intensive irrigation systems as drip and sprinkler for the production of fruits, vegetables and the ornamental crops. The fruits include mainly mango, papaya and pineapple. The vegetables include mainly the peppers, okra and garden eggs. There is hardly any statistics on the land area covered by this form of irrigation but could be over 1,000 ha.

1.5 The Irrigation System

The typical irrigation system comprises:

a) Intake structure or pump house connected to a water source
b) Conveyance and distribution system
c) Field water application system
d) Drainage system

Intake structure or pump house connected to a water source: The intake structure is built at the entry to the irrigation system. The purpose is to direct water from the original source of supply into the irrigation system. Where the irrigation water source is at a higher level than the level of the farm, the irrigation

system is fed by gravity from the source. Where the irrigation water source lies below the level of the irrigated fields, a pump must be used to supply water to the irrigation system. The water source can be surface water or groundwater. Surface water can be a river, natural or artificial lake (dam). Groundwater can be accessed by wells, boreholes and wash bores. There can be direct abstraction of water from rivers and lakes. In several cases, artificial reservoirs or lakes are created by damming streams or rivers. This is done to even-out the floods and the droughts by storing water when there is more than enough and using it when there is less than enough.

Conveyance and distribution system: The conveyance and distribution systems consist of canals or pipes transporting the water through the whole irrigation system.

Canals are open channels dug in the ground, lined or unlined and carry water from one place to another. The lining involves placing hard and/or impermeable materials on the canal surfaces to stabilize the structure and to prevent it from caving in due to erosion. The lining material can be concrete mixtures, geotextiles or even stable, cohesive soil materials. Unlined canals are preferred in stable, cohesive soils with high resistance to erosion by raindrop impact and water transport. Otherwise, the canal becomes easily eroded and gets damaged. According to the shape of their cross-section, canals are called rectangular, trapezoidal, circular, parabolic and irregular/natural. The most commonly used canal cross-section in irrigation and drainage is the trapezoidal cross-section because this is considered to be the most stable due to the sloping sides. Canals are constructed with a bed slope to enable water to flow by gravity and the velocity of flow in the channel depends on the bed slope. Open channel hydraulics will be discussed in more detail later.

Canal structures are required for the control and measurement of water flow. The types, functions and examples of canal structures are shown in Table 1.3.

Table 1.3: Canal structures

Type of canal structure	Function	Examples
Distribution control structure	Controls flow distribution and directs water to different places on the field	Division boxes, checks, turn-out
Erosion control structure	Controls erosion in the channel carrying the water by reducing flow velocity or improving channel lining	Drop structures, chutes
Crossing structure	Carries water across important land marks without interfering in the use of such land marks	Culvert, inverted siphon, flumes
Water measurement structure	Measures flow in the canal as a means of monitoring water delivery efficiency	Parshall flume, cut-throat flume, weirs

Pipes are closed channels which are manufactured in short sections and can be joined together to any length using couplings. Pipes are available in different sizes and are made of different materials but the commonest are the PVC pipes. They are manufactured to withstand different pressures and where such pressures are exceeded without any mechanism for pressure release, the pipe may burst. Flow in pipes result in friction head losses that occur at the boundary between the moving liquid and the pipe material.

There are pipe accessories or fittings that enable the installation of a network of pipes. These include hydrants, couplings, Ts, Ys, elbows, expanders and reducers, pressure gauges, water meters, pressure relief valves and other valves for flow control (Table 1.4).

Table 1.4 Pipe accessories/fittings

Type of accessory/fitting	**Function**
Hydrant	Water take-off point, allowing pipe connections for the delivery of water to the field
Couplings	Enable pipes to be joined to any length. In order to make them water tight, they are made with seals
Ts, Ys, Elbows	Water distribution, allowing re-direction of water to any part of the field
Expanders and Reducers	Allow, where necessary, the changing of the size of pipe from a small to a larger size (expander) and from a large to a smaller size (reducer)
Pressure gauge	Enables the pressure in the pipe to be measured or monitored
Water meters	Measures flow in the pipe as a means of monitoring water delivery
Pressure relief valve	Acts as a safeguard on pipes by releasing excess pressure that builds up in the pipe
Valves for flow control	Enables flow in the pipe to be controlled by opening and closing the valves

Any good fluid mechanics or engineering hydraulics text can provide further details on pipe hydraulics. Closed channel hydraulics will be discussed in more detail later.

Field Water Application System: There are many methods of applying water to the field. The simplest is fetching water with a bucket or watering can from a source and applying it manually to the field. This can be time-consuming, less efficient and can be applied to only small fields. More sophisticated methods of water application are used in larger irrigation fields. There are 3 basic methods: surface, sprinkler and drip irrigation.

Surface irrigation: This involves flooding the field from a head ditch through furrows, borders and basins. Requiring reconstruction of the irrigation field, a slight slope in the field allows for water applied at one end of the field to advance to the other end. While the water advances on the surface, it infiltrates into the soil as well such that the upper end of the field receives more water than the lower end of the field. This non-uniform application of water suggests that for all parts of the field to receive adequate amounts of water into the soil, more water must be applied, thus resulting in excessive runoff and deep percolation losses. To minimise these losses, irrigation stream flow is applied intermittently and not continuously. Irrigation efficiencies are usually low for this type of field application method with typical values between 40 to 55 %.

Sprinkler irrigation: In this case, artificial rainfall is created. There is no need to reconstruct the field as in the case of surface irrigation. The water is carried through a pipe system in which the water is under pressure. The spraying of water in the form of rain is accomplished by using several rotating sprinkler heads or spray nozzles or a single gun type sprinkler. Typical efficiencies achieved are between 70 and 80 %.

Drip irrigation: Also known as trickle irrigation, the water is carried to the field through a pipe system under pressure but requires an extensive network of pipes called laterals that lead to every plant in the field. These lateral pipes are fitted with drippers or emitters to enable water application to be done slowly, drop by drop near the plants. This system lends itself to chemigation which is the combined application of irrigation and agrochemicals I one process. Because the drippers have the tendency to be blocked, the system has primary and secondary filters that remove most suspended solids in the water that may block the dripper from flowing. It is the most capital extensive but the most efficient of all the irrigation methods. Used mainly for high value crops such as fruits, nuts and vegetables, irrigation efficiencies can be 90 % and above. This system also has a limitation of not having sufficient soil moisture buffer as in the case of the surface and sprinkler methods.

Drainage System: An irrigation system must be complemented by a field drainage system that can manage excess water that may arise from over-irrigation and excessive rainfall. Also, due to the intensive nature of cultivation, continuous application of agrochemicals may result in salt accumulation in the soil. This salt build-up can result in the loss of the productive potential of the land. Dealing with this issue will require leaching the root zone with large amounts of irrigation water and getting rid of the resultant poor drainage water through field drains. Quite often, the natural drain channels are used as the storm drains in case of excessive rainfall. Drainage systems can be surface or subsurface. The surface drains remove excess water from the surface of the field and these are achieved through open surface channels. Subsurface drains remove excess water and salts from the soil. These are achieved by deep open drains and pipe drains installed underneath the surface.

1.6 Exercises

1. Briefly define irrigation and explain why it is important.
2. Irrigation is not a substitute for rainfall but rather a supplement to rainfall. Explain.
3. Classify irrigation in accordance with a) rainwater availability and b) water delivery.
4. What is the water year in Ghana? Briefly explain the hydrology and the agroecology of Ghana, linking it to irrigation practice in Ghana.
5. What government agency is responsible for irrigation in Ghana? In the context of irrigation policy, classify irrigation practice in Ghana, briefly explaining each classification.
6. List the components of a typical irrigation system and briefly explain in not more than a sentence each, the function of each component.
7. Name the canal structures you know and briefly explain their functions with examples.
8. List 4 accessories of a pipe network system and briefly explain their functions.
9. Name the irrigation methods there are, stating the most efficient of them with reasons.
10. Explain why it is important to have drainage systems in irrigation fields.

Module 2: Soil-Plant-Water Relations

2. 1 **Introduction**

The purpose of irrigation is agriculture. Plants are grown in the soil but technology has made it possible to grow plants in other soil-less media, to the extent that providing the right rooting conditions, plants can be grown in water (hydroponics). However, the soil will continue to be one most important resource that mankind will continue to depend on for the sustainability of life. This module explains the soil as a water reservoir for plant growth and the procedures for estimating the irrigation related parameters.

2.2 **The Soil as a Water Reservoir**

A soil is considered to be any loose sedimentary deposit such as gravel, sand, silt or clay or a mixture of these materials. The characteristics of the topsoil are of importance in agriculture. Usually the top soil contains organic matter which decomposes with respect to time and changes the behavior of the soil. The individual particles which may vary considerably in size and shape form a framework of solid material and between the individual particles is a system of interconnected spaces. The framework of soil particles is referred to as the soil structure and the spaces between particles as voids or pores.

Below the water table, the pores will generally be completely filled with water while above the water table, the pores will partly be filled with water and air. When the pores are completely filled with water, the soil is said to be saturated. If they are filled with both water and air, the soil is said to be unsaturated. If the pores are completely filled with air, the soil is said to be air-dry. Therefore, strictly speaking, all soils are said to be not only porous but permeable.

For agronomic purposes, the water stored in the pore spaces are extracted by plant roots and the soil volume explored is dependent on the availability of soil water. Further discussion on this will be in the subsequent sections.

2.3 **The Physical Properties of the Soil**

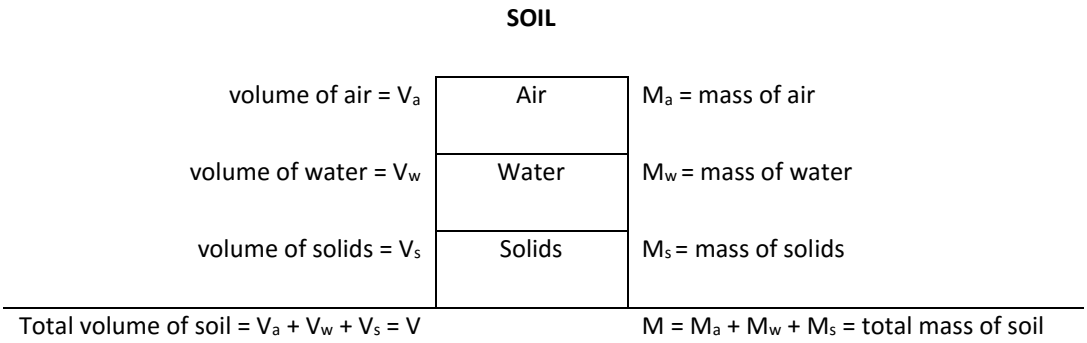


Figure 2.1: Schematic diagram of a soil mass

From the diagram above,

Soil Water Content

a) Mass water content of the soil (θ_m) is defined as:

$$\theta_m = \frac{M_{\mathrm{w}}}{M_s} \tag{2.1}$$

b) Water content by volume (θ_v) is defined as:

$$\theta_v = \frac{V_{\mathrm{w}}}{V} \tag{2.2}$$

Soil Density

c) Soil wet bulk density (ρ_{wb}):

$$\rho_{wb} = \frac{M}{V};\ \text{where } M = M_{\mathrm{w}} + M_{\mathrm{s}};\ V_w < V_p \tag{2.3}$$

d) Soil dry bulk density (ρ_d):

$$\rho_d = \frac{M_{\mathrm{s}}}{V} \tag{2.4}$$

e) Soil particle density (ρ_s):

$$\rho_s = \frac{M_{\mathrm{s}}}{V_s};\ \text{soil particle density is about } 2.65\ \mathrm{g/cm^3} \text{ for all soils.} \tag{2.5}$$

f) Soil saturated density (ρ_{sat}):

$$\rho_{sat} = \frac{M}{V};\ \text{where } M = M_{\mathrm{w}} + M_{\mathrm{s}};\ V_w = V_p \tag{2.6}$$

g) Submerged density (ρ_{sub}):

$$\rho_{sub} = \rho_{sat} - \rho_w \tag{2.8}$$

Soil porosity, void ratio, degree of saturation and air voids content

h) Porosity (p):

$$p = \frac{V_{\mathrm{v}}}{V} \tag{2.9}$$

i) Void ratio (e):

$$e = \frac{V_v}{V_s} \quad (2.10)$$

j) Degree of saturation (θ_s):

$$\theta_s = \frac{V_w}{V_v}; \text{ If } \theta_s = 1; \textit{the soil is saturated};\ \theta_s = 0; \textit{the soil is air dry} \quad (2.11)$$

k) Air voids ratio (θ_a)

$$\theta_a = \frac{V_a}{V} \quad (2.12)$$

Quantity of water present in the soil

l) Depth of water (d) present in a certain depth (profile) of soil (Z):

$$d = \theta_v Z \quad (2.13)$$

m) Volume of water (V_w) present in the soil over an area (A):

$$V_w = dA \quad (2.14)$$

For the purposes of irrigation, the concepts *I* and *m* are the most important.

Relationships between soil physical properties

a) $\theta_v = \rho_d \theta_m$ θ_m = gravimetric water content (2.15)

b) $\theta_a = p - \theta_v$ θ_v = volumetric water content (2.16)

c) $\theta_a = p\,(1 - \theta_s)$ θ_s = degree of saturation (2.17)

d) $\theta_a = \frac{e - G_s\theta_m}{1 + e}$ θ_a = air voids content (2.18)

e) $\rho_{wb} = \rho_d(1 + \theta_m)$ ρ_d = dry bulk density (2.19)

f) $\rho_d = \frac{G_s\rho_w}{1 + e}$ ρ_{wb} = wet bulk density (2.20)

g) $$\rho_d = \frac{G_s \rho_w (1 - \theta_a)}{1 + G_s \theta_m}$$ (2.21)

ρ_w = density of water

G_s = specific gravity of soil particles

h) $$p = \frac{e}{1 + e}$$ (2.22)

p = porosity

i) $$\theta_s = \frac{\theta_m G_s}{e}$$ (2.23)

e = void ratio

2.4 **Soil Water Capacity**

The soil stores water for plant use and the size of the soil water reservoir also known as soil water capacity depends on the soil type. The coarse textured soils have smaller soil water capacities while the fine-textured soils have larger soil water capacities. This leads to the explanation of the subsequent concepts.

Soil saturated water content (θ_{sat}): This is the water content of the soil at saturation or when the soil pores are completely filled with water. This water quickly drains by gravity from the soil such that it is not available for plant use. The soil water energy level at saturation is deemed to be 0 bar.

Field capacity water content (θ_{fc}): This is the most ideal water content of the soil for plant growth. It is the water content of the soil 24 to 48 hours after full saturation of the soil usually by rainfall or irrigation. The soil water potential at field capacity is said to be 0.1 or 0.3 bar.

Drainable porosity (θ_{dp}): This is the difference between saturation and field capacity water contents. This is not available for plant use. It is expressed mathematically as:

$$\theta_{dp} = \theta_{sat} - \theta_{fc} \quad (2.24)$$

Permanent wilting point water content (θ_{pwp}): This is the water content of the soil at which the plant will permanently wilt and die. This water is held at -15 bars. This is different from temporary wilting.

Soil total available water capacity (TAWC): This is the total amount of water retained in the soil and available for plant use. It is the difference between field capacity and permanent wilting point water contents. This is expressed mathematically as:

$$TAWC = \theta_{fc} - \theta_{pwp} \quad (2.25)$$

Readily available soil water capacity (RAWC): This is a fraction of the *TAWC* and it is that water which is readily available for plant use. This fraction is known as the soil water depletion factor (p). Since plants experience more water stress as the water content decreases, it is advisable not to deplete the TAWC before more water is added. Same as the net irrigation requirements, it is expressed mathematically as:

$$RAWC = p\,TAWC \quad (2.26)$$

Soil water depletion factor (p): This is the ratio between the *RAWC* and the *TAWC*. It is the maximum allowable fraction of the *TAWC* before irrigation is applied again. Assuming the soil water content at which irrigation is applied is denoted as θ_i the soil water depletion factor can be expressed mathematically as:

$$p = \frac{\theta_{fc} - \theta_i}{\theta_{fc} - \theta_{pwp}} \quad (2.27)$$

Hygroscopic water: This is the amount of water retained in the soil beyond the permanent wilting point water content. This water is so tightly held in the soil such that it cannot be used by the plant.

2.5 **Soil Water Potential**

Soil water is said to be held at certain energy levels and requires the plant to do some work in extracting the water. It is referred to as soil water pressure, potential or suction. It is a negative pressure. These soil water energy concepts are explained subsequently.

Total soil water potential (*H*): This is the total pressure under which the soil water is held at any given location. It is the sum of matric (Φ_m), gravitational (Φ_g) and osmotic potentials (Φ_o).

$$H = \emptyset_m + \emptyset_g + \emptyset_o \quad (2.28)$$

Soil matric potential (Φ_m): This is the pressure of water in the soil matrix or pore.

Soil gravitational potential (Φ_g): This is the pressure of water in the soil with respect to its position within the gravitational field.

Soil osmotic potential (Φ_o): This is the pressure of water in the soil with respect to salt accumulation in the soil. Salt accumulation problems can be either salinity or sodicity. Instead of water flowing from the soil into the roots, water rather moves from the roots to the soil.

Soil hydraulic potential (Φ_h): This is the sum of the matric and gravitational potentials.

$$\emptyset_h = \emptyset_m + \emptyset_g \quad (2.29)$$

Soil water retention or characteristics curve: This is the relationship between soil matric potential expressed as the *pF* and the soil water content. It expresses the behavior of the water retention properties of the soil. The pF is the negative log of the soil matric potential, expressed as:

$$pF = -log_{10}(\emptyset_m) \quad (2.30)$$

2.6 **Soil Infiltration, Permeability and Deep percolation**

Soil infiltration, permeability and percolation are important parameters that are used to design and manage irrigation systems. These concepts will be explained as follows:

Soil infiltration: The penetration of water into the soil from the soil surface.

Soil infiltration rate: The rate of infiltration of water into the soil. There is an initial rate of infiltration which occurs at the onset of the application of water. The final or basic infiltration rate is the practically constant rate of infiltration of water into wetted soil after time. The Kostiakov equation expresses the infiltration rate (*i*) as a function of time *(t*) as:

$$i = kt^n \quad (2.31)$$

where, *k* and *n* are empirical constants. The Philips equation relating *i* as a function of *t* is also expressed as:

$$i = St^{-1/2} + K_{sat} \quad (2.32)$$

where, *S* is the soil sorptivity and K_{sat}, the soil saturated hydraulic conductivity

Soil saturated hydraulic conductivity: The ability of the soil to transmit water from one location to another depending on a hydraulic gradient. The flow velocity (*V*) is related to the K_{sat} in the Darcy equation as:

$$V = -K_{sat}\frac{\partial h}{\partial l} \quad (2.33)$$

where, $\frac{\partial h}{\partial l}$ is the hydraulic gradient.

Deep percolation: The portion of rainfall and irrigation that percolates through the root zone of the crop before it can be utilised by the crop.

2.7 **Precipitation and Evapotranspiration**

Precipitation and evapotranspiration are two important components of the water cycle and the important parameters on which irrigation design and management computations are based. While precipitation represents the supply side of water to the plant, evapotranspiration expresses the demand side of water by the crop.

Rainfall or Precipitation (*P*): The depth of precipitation registered during a given period by a correctly mounted rain gauge over a given space. Amount, duration and areal extent are the important parameters in rainfall measurements. Rainfall intensity is the ratio of rainfall amount to duration within which it falls. It is measured in mm/h. A rain gauge is considered to have an area of influence and the records of the network of gauges can be used to determine the average rainfall over the total space covered by the network. Average rainfall over space can be determined by the a) Arithmetic Mean Method; b) Theissen Polygon Method; and c) the Isohyetal Method.

Reliable Rainfall (P_r): That amount of the rain that is certain to fall 75 or 80 % of the time in a certain space and in a given period. In other words, it is that rain that is certain to fall 3 out of every four years or 4 out of every five years over a given space. Using the Weibull method rainfall frequency analysis, rainfall frequency (*f*) of occurrence can be computed as:

$$f = \frac{100\,m}{n+1} \quad (2.34)$$

where, *m* and *n* are rank number and number of years of rainfall data respectively. For project planning and design, it is recommended that at least 30 years of continuous data is needed for this analysis.

However, in the absence of that, some data is better than no data and shorter periods of at least 10 years will be acceptable.

Effective rainfall (P_e): The rain that falls during a given cycle which can be used profitably by the plants and that partially or completely replaces scheduled irrigation. It is a certain percentage of the total rainfall. The concept of effective rainfall varies depending on usage. The definition above reflects the agronomist's concept. The hydrologist's view is the portion of rain that constitutes runoff into the surface waters while the hydro geologist considers it as that portion of rain that goes beyond the root zone to recharge the groundwater table. Effective rainfall can be expressed as:

$$P_e = rP \tag{2.35}$$

It can also be expressed as:

$$P_e = f(D)[1.25P^{0.824} - 2.93]10^{0.00095ET_c} \tag{2.36}$$

$$f(D) = 0.53 + 0.0166D - \frac{8.94D^2}{10^5} + \frac{2.32D^3}{10^7} \tag{2.37}$$

$$D = 10\left(\theta_{fc} - \theta_{pwp}\right)p\rho_d Z \tag{2.38}$$

$$ET_c = k_c ET_o \tag{2.39}$$

where, ET_c = crop evapotranspiration (mm); P_e = effective rainfall (mm); k_c = crop factor; ET_o = potential evapotranspiration (mm); P = reliable rainfall (mm); D = normal depth of soil water prior to irrigation (mm); θ_{fc} = field capacity water content of soil (% wt.); θ_{pwp} = permanent wilting point water content (%wt.); r = fraction of rain used by plants; p = soil water depletion factor; ρ_d = dry bulk density of soil; Z = average rooting depth of soil (m).

Evaporation (E): The loss of water from the soil surface. Evaporation can be weather-controlled (constant rate) or soil-profile controlled (falling rate). It forms part of evapotranspiration.

Transpiration (T): The loss of water into the atmosphere from the growing plant regulated by physical and physiological processes in the plant. It forms part of evapotranspiration

Crop Evapotranspiration (ET_c): The loss of water from the plant and its environment. It is the sum of transpiration (T) and soil evaporation (E).

Reference/Potential Evapotranspitation (ET_o): Evapotranspiration of a reference crop (green grass of uniform height 80 to 150 mm tall), growing actively, completely shading the ground and not short of water.

Pan Evaporation (E_{pan}): The amount of water that evaporates from an open water surface in an evaporation pan during a certain period.

Pan coefficient (k_{pan}): The ratio of reference evapotranspiration (ETo) to pan evaporation ($Epan$)

$$ET_o = k_{pan}E_{pan} \tag{2.40}$$

2.8 **Determination of Evapotranspiration**

Measurement of evapotraspiration can be done by direct measurement or indirectly using empirical formulae. The direct measurements include lysimetry and the use of evaporation pans. Conventional lysimeters are bulky and expensive to install but there are smaller versions as well. For the purposes of this text, emphasis will be placed on the evaporation pan.

Evaporation pan: This is a device used for the measurement of the amount of water lost through evaporation from an open water surface during a specific period. An example is the Class A pan. The class A pan has the following features:

- Dimension: 120.7 cm diameter; 25 cm (250 mm) depth;
- Maximum level of water in pan is 20 cm (200 mm);
- Mounted on wooden platform for maximum air circulation;
- Readings: daily readings in the mornings (9.00 hrs GMT) and can be done in 2 ways: 1) refilling the pan to the maximum level or reference level; 2) adjusting a pointer fitted to a vernier scale to the level of the water in the pan.

The class A pan can be used to schedule irrigation. The *Epan* measurements using the depth gauge is expressed as:

$$E_{pan} = d_i + P - d_{i-1} \tag{2.41}$$

where, E_{pan} = pan evaporation (mm); d_i = present day gauge reading (mm); P = rainfall since last reading (mm); d_{i-1} = previous day gauge reading (mm)

Pan evaporation measurements based on refilling pan to reference level:

$$E_{pan} = \frac{40V}{\pi d^2} \tag{2.42}$$

where, E_{pan} = pan evaporation (mm); V = present day gauge reading (cm^3); d = diameter of class A pan (cm)

Empirical Formulae: A number of equations were developed based on climatic data for the estimation of potential evapotranspiration. They have their advantages and limitations but the most widely used as the standard method is the Penman-Monteith equation. This equation assumes that evapotranspiration occurs through the combined effect of solar radiation and wind effect and uses climatic data as temperature, relative humidity, wind velocity, sunshine duration and solar radiation as input for the computation. The simplest form of this equation is given as:

$$ET_o = c[wR_n + (1-w)(e_s - e_a)f(u)] \tag{2.43}$$

$$c = f(RH_{max}, \frac{U_{day}}{U_{night}}, R_s, U_{2d}) \tag{2.44}$$

$$w = \frac{\Delta}{\Delta+\gamma} \quad (2.45)$$

$$f(u) = m(x_1 + x_2 U_2) \quad (2.46)$$

where, c = calibration coefficient based on meteorological data; w = weighting function for elevation and temperature; R_n = net radiation (equivalent mm/d); e_s=saturation vapour pressure at air temperature (mb); e_a = mean actual vapour pressure of air (mb); $f(u)$ = wind function; R_s = solar radiation (mm/d); RHmax = mean maximum relative humidity (%); U_{day}/U_{night} = day to night wind run ratio (daytime arbitrarily chosen as 0700 – 1900 hrs); U_{2d} = daytime wind speed at 2 m height (m/s); Δ = slope of saturation vapour pressure versus temperature curve at Tmean; γ = psychrometric constant (relationship between vapour pressure deficit and wet bulb depression); m = conversion constant; x_1, x_2 = empirical constants; U_2 = mean daily wind run at 2 m height (km).

2.9 **Determination of Irrigation Requirements**

Strictly speaking, the *RAWC* is equivalent to the **net irrigation water requirements**. It is the net amount of water extracted by the plant from the soil to meet its requirements within an irrigation cycle. Expressed in depth units of water over an average rooting depth, the net irrigation requirements (I_n) can be expressed as:

$$I_n = 10(\theta_{fc} - \theta_{pwp})p\rho_d Z \quad (2.47)$$

Gross irrigation water requirement (I_G) is the net irrigation plus the losses due to system inefficiencies. For field application efficiency (E_a), the gross irrigation requirement is expressed as:

$$I_G = \frac{I_n}{E_a} \quad (2.48)$$

Irrigation interval (*II*): This is the irrigation cycle and equivalent to the number of days it takes a plant, depleting water at a rate to exhaust the net irrigation amount. It is expressed mathematically as:

$$II = \frac{I_n}{ET_c} \quad (2.49)$$

2.10 **Irrigation system efficiency (E_p)**

Efficiency is the output of a specific operation in relation to the input. Irrigation efficiency is water actually delivered for use by the plant in relation to water supplied. It includes conveyance efficiency (E_c), field distribution efficiency (E_d) and field water application efficiency (E_a)

Conveyance efficiency (E_c): This is the total water delivered to the farm gate or draw-off point in relation to the total water transported from the source.
Field water distribution efficiency (E_d): This is the total water applied to the soil surface in relation to the total water delivered to the farm gate or draw-off point.

Field water application efficiency (E_a): This is the total irrigation water in the root zone of the soil used by the plant in relation to total water applied on the soil surface.

The project irrigation efficiency is therefore given as:

$$E_p = E_c E_d E_a \tag{2.50}$$

2.11 **Exercises**

1. Fill in the blank spaces below.

   a) The 3 important rainfall parameters often measured include _____ _____
   b) Rainfall intensity is expressed as a ratio between _____ and _____
   c) Averaging rainfall over space can be done using three (3) methods, namely: 1)_____, 2)_____ and 3)_____.
   d) Evapotranspiration is the sum of _____ and _____.
   e) Constant rate soil evaporation is also known as _____ evaporation.
   f) Five millimeters of evaporation from the surface of a 54.5 ha irrigation reservoir is equivalent to _____ m^3 of water.
   g) Soil sample taken from an irrigated farm was found to have 1.45 g/cm^3 wet bulk density at 18.6 % gravimetric water content. The volumetric water content of the soil would be_____ %.
   h) The gross irrigation required at an irrigation efficiency of 70 % for a sprinkler system was 100 mm. The irrigation interval based on peak crop water demand of 7.0 mm/d will be _____ days.
   i) The amount of irrigation water applied on a 2.5 ha farm for a crop water requirement of 75 mm at a field water application efficiency of 75 % is _____ m^3.

2. A soil with a readily available soil water capacity on weight basis amounts to 5.5 %. If the soil has a relative dry density of 1.3, what will be the net irrigation requirement for an average rooting depth of 1.2 m?
3. If the field capacity water content for the soil in *question 2* is 20.0 % (weight basis), what is the water content at irrigation?

Problem: The soil of an irrigation farm has field capacity and permanent wilting point water contents of 25.5 % and 11.3 % (weight basis) respectively. The relative soil density = 1.2; average rooting depth = 1.5 m; soil moisture depletion factor = 0.45 irrigation application efficiency = 65%; peak crop evapotranspiration = 7.67 mm/d. ***Answer the next 3 questions.***

4. What is the net irrigation requirement?

5. What is the gross irrigation requirement?

6. What is the design irrigation interval?

7. Compute the net irrigation required for a soil with the total available soil water capacity of 0.152 (volume basis) over a rooting depth of 1.0 m and for a soil moisture depletion factor of 0.5. The dry bulk density of the soil is 1.32 g/cm^3.

8. Soil samples were taken at different locations from a 0.4 ha farm using the core samplers (10 cm diameter x 5 cm length) about 48 hours after a rainfall event. The following measurements were made before and after oven drying the samples:

Sample No.	Mass of wet sample (gm)	Mass of dry sample (gm)
1	651.49	549.78
2	644.90	541.93
3	632.47	530.15
4	613.12	510.51
5	647.92	545.85
6	628.83	526.22
7	593.96	490.88
8	602.19	502.66
9	572.56	471.24
10	644.44	542.00

a) Calculate the average water content of the soil (% volume).
b) Calculate the average depth (mm) of water present in 50 cm depth of the soil.
c) Calculate the average volume of water present in the soil.
d) Assuming the soil water conditions at the time of sampling represent the field capacity water content of the soil and assuming further that the average permanent wilting point water content of the soil on the farm is 8.73 % by mass, compute the average water content of the soil at irrigation for an allowable soil water depletion factor and average rooting depth of 0.55 and 1.2 m respectively.
e) Calculate the average net irrigation requirement (mm) on the farm.
f) Calculate the gross amount of irrigation required (mm) if irrigation is applied at 70 % efficiency.

Module 3: Feasibility Study for the Design of Irrigation Systems

3.1 **Introduction**

Irrigation system design always starts with the identification of a suitable irrigation site and the collection of relevant field data that will make it possible for the formulation of the design of the project. This module will explain the procedures for gathering relevant and sufficient data and also the type of data required.

3.2 **Feasibility Study**

The aim of undertaking feasibility studies is to do site selection for the irrigation scheme based of reliability of water source, suitability of land and accessibility and to collect and analyse data necessary for the design of the scheme. It involves 1) pre-feasibility study and 2) detailed feasibility study.

Pre-feasibility study: This involves desk study, field visit for reconnaissance survey and site selection. Desk study normally involves the study of maps acquired from the national survey department (cartography, air photo, satellite imagery). Digital maps may be obtained from the internet. Maps are examined for drainage features, topography of land, soil and vegetation type and best access route to possible site. The desk study is followed by a field visit for reconnaissance survey and selection of possible project site. The field visit requires a team of experts in the areas of irrigation, hydrology, soils and crops, socioeconomy and environmental science. Reconnaissance survey of site is done through transect walking to establish reliability of water source and suitability of soil. It requires also interacting with the community nearest to site to verify climate trends, water availability, soil suitability, demography, livelihood indicators, social services, market access and environmental issues. Based on a combined view of the team of experts, site selection for the project is done.

Detailed feasibility study: The aim here is gather and analyse information relevant for scheme design and management. The detailed field study begins with the preparation of a study plan detailing the number of experts for the study, number of days required for the field study, logistics (vehicle and consumables) and the budget. Data required include climate, soil and topography, crop, hydrology, socioeconomy and environment data specific to the site.

Climatic data required include 30-year average data for precipitation, evaporation, temperature, relative humidity, wind velocity, sunshine duration and solar radiation. Quite often, climatic data may not be available for the specific site but data for the nearest meteorological station can be used. It is also possible that the 30-year data may not be available but some data is better than no data at all. In such a situation, at least a 10-year record can be used. Climatic data can be obtained from the national meteorological offices.

Soil and topographic data required can be grouped into: 1) system design soil data and 2) agricultural suitability soil data and 3) topo-survey data. The system design soil data include field capacity water content, permanent wilting point, soil infiltration rate, soil saturated hydraulic conductivity, electrical conductivity and sodium adsorption ration (SAR). Soil data for agricultural suitability may include but not limited to texture, drainage, carbon content, nitrogen content, C/N ratio, cation exchange capacity (CEC), and potassium and phosphorus contents. The soil data can be obtained with the help of soil scientists.

Finally, a topo-survey map for the entire scheme area, detailing drainage features, important land marks and contours is done with the help of a surveyor or geodetic engineer. Minimum level of the source of water, especially for the surface sources should be indicated on the contour map. Additional soil data can be obtained from national soils offices.

Crop data for scheme design and management include average rooting depth of crop, crop coefficient, growth period. Crop water requirements which is computed as crop evapotranspiration can be computed from climatic data and crop coefficient. Other crop husbandry data can also be obtained. Additional data can be obtained from the national agricultural research stations.

Hydrology data considers surface and groundwater sources. Surface water sources rivers, natural and artificial lakes created by dams and weirs. For dams and weirs, peak flood estimates are required. For direct abstraction from rivers, minimum discharge and water level estimates are required. Whereas measured stream flow data is preferred for establishing stage-discharge (rating) curves for the streams, empirical methods can also be used to predict stream flow data. Thirty (30)-year measured hydrological data are required but such data are quite often not available and hence shorter periods can be considered. For lakes, stage-capacity curves are required to be able to estimate the supply levels. Ground water data required include minimum aquifer yield and groundwater level. These can be obtained from secondary hydrogeological data previously obtained in well drilling activities. If a well ought to be drilled as the supply source, pumping test data as drawdown and recovery data are needed to estimate aquifer properties of transmissivity and storage coefficient, and well yield. National hydrological services offices and water research institutes can provide additional data for the study.

Socioeconomic data deals with demography, social services, poverty indicators, market access, and prices of goods and services. The demographic data should include but not limited to population, density and growth rate, educational background, religion, employment and income level. Social services data must include access to education, school enrolment and dropout rates, health facilities, infant and maternal mortality rates, water and sanitation facilities, roads, transport and telecommunication facilities, market access and, prices of goods and services. Benefit-cost analysis is usually done to determine the economic impact of the project based on with or without project scenario analyses. Additional data sources can be obtained from the national statistical services offices.

Environmental data is required to assess the positive and negative impacts of the irrigation project. Assessment should be done on such environmental degradation indicators as climate change, bushfires, loss of soil fertility, vegetation loss rate or rate of deforestation and, loss of biodiversity. An environmental impact assessment (EIA) should be carried out to weigh the positive and negative effects of the project. This exercise is fairly subjective but provides basis for the determination of the environmental soundness or otherwise of the project. National environmental offices can be contacted for additional information.

3.3 **Design and Costing of the Project**

Once relevant data has been gathered, the next step is largely a desk study requiring design computations and design drawings. Where the project is socially, economically and environmentally justified, cost estimates are done and a complete design portfolio is prepared for tender. Subsequent chapters will deal with the design considerations for the various components of the irrigation system.

3.4 **Exercises**

1. What is the aim of feasibility study prior to the design of irrigation projects?
2. Briefly explain how the selection of the irrigation project site is done.
3. Briefly explain the steps involved in feasibility study of an irrigation project.
4. List and briefly explain the data required in the design of an irrigation project.

Module 4: Water Resources Development

4.1 **Introduction**

Irrigation is a water development activity. It requires matching the demand of water by the plants to the water supply levels, making sure that even in the extreme of conditions, there will be sufficient amounts of water to meet the crop water requirements. The water sources include surface and groundwater sources. The surface sources are natural and artificial lakes created by dams and weirs, perennial streams and rivers. Where groundwater sources are to be used, wells are developed and abstractions made by pumping water to the surface to irrigate farms. The subsequent sections of this chapter will deal with the design of small dams and wells.

4.2 **Design of Small Dams**

There are many types of dams, serving purposes of hydropower generation, water supply, flood control and irrigation. The design and construction of dams and weirs require engineering skills. Small earth dams will be considered in this chapter. A typical dam (Figure 4.1) has two main parts: the embankment and the reservoir. The typical features are as follows:

Embankment	**Reservoir**
• Height of dam • Length of embankment • Crest width • Upstream and downstream slopes • Cut-off trench • Impermeable core • Toe drain • Spillway • Stone pitching/rip rap on upstream slope • Grass cover on downstream slope	• Maximum reservoir level • Maximum depth of water in the reservoir • Maximum reservoir capacity • Maximum surface area of the reservoir • Maximum throw-back length of reservoir • Total live storage • Dead storage level

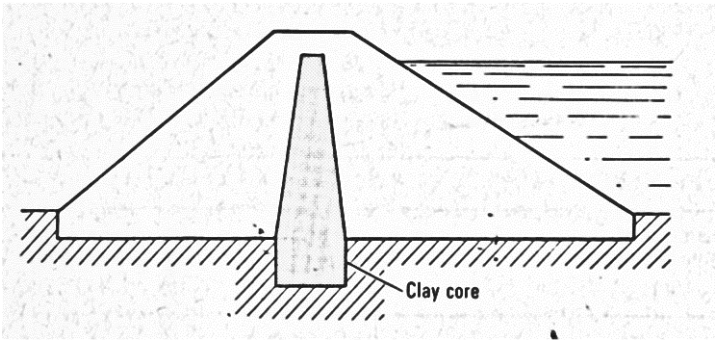


Figure 4.1 Typical cross-section of an earth dam

Planning and site location of dam: Defining clearly the purpose for the dam is the first step in planning so that design requirements can be identified. If the purpose is irrigation, the amount of water storage and the period the water must be available should be defined. Also, the mode of water delivery should

determine the location of the dam. If water delivery is by gravity, the dam must be located upstream and sufficiently above the level of the farm downstream. If the topographic effects will not allow gravity flow, then the dam can be located at a place where water is delivered to the farm by pumping.

Siting the dam should be considered on the basis of reservoir storage, spillway and soil. For the **reservoir storage**, the shape of the valley influences how much water will be stored for a given height of dam wall. The ideal site should have steep valley sides at the dam site so the embankment is short, long reservoir throw-back upstream giving a big capacity and a small reservoir surface area. A valley with gently sloping sides means a long and expensive wall and consequently a large reservoir surface area at relatively shallow depth. Smaller reservoir surface area means lower evaporation and vice versa. Spillways serve as pressure relief valves for dams in the event of flush floods allowing for water passage when the reservoir is full. Building concrete spillways add to the cost of small dams but selecting sites where the overflow can be safely discharged over grass spillways reduce the cost appreciably. The size of spillways depends on the maximum expected flood and this in turn depends on the size of the catchment area and the rainfall. Grass spillways should be constructed only if the reservoir catchment area is up to 250 ha and mean annual rainfall more than 625 mm. Where the mean annual rainfall is less than 625 mm, grass waterways can be constructed if the reservoir catchment area is up to 500 ha.

The site must have suitable soil conditions for constructing an earth dam. Porous soils as sandy soils and sites with boulders and termite mounds are not suitable for basin conditions due to excessive seepage that may occur. Unsuitable soils for dam construction include saline, alkaline/sodic soils, soils with high organic matter content, peat soils, and heavy clays with swelling and shrinkage properties, light sands and soils with high proportion of fine silt. Suitable soils needed for construction should not be far from the construction site in order to cut down on haulage costs. Construction materials should be about 1.5 times the total compacted volume of embankment.

Estimating the embankment volume: Quantity of earthworks (Figure 4.2) can be estimated by methods of vertical and horizontal slices. The approximate formula for estimating the dam volume is:

$$V_e = 0.216HL(2w + HS) \tag{4.1}$$

Where,
V_e= volume of dam (m^3); $H =$ the maximum height of wall (m); $L =$ maximum length of dam wall along the crest (m); $w =$ the crest width (m); $S =$ sum of the upstream and downstream slopes of the embankment.

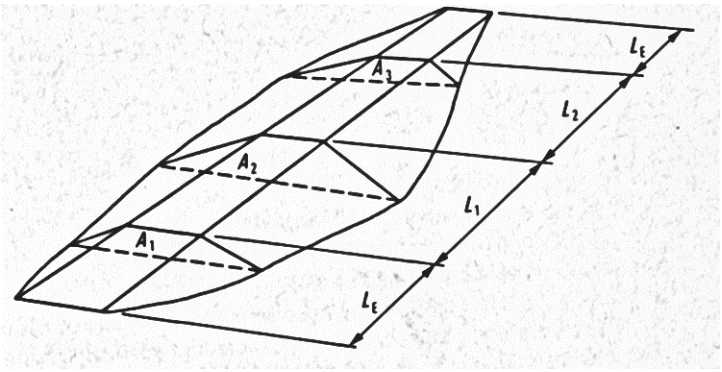


Figure 4.2 Method of vertical slices for estimating volume of dam

More accurate estimates can be obtained by considering the embankment as a series of slices and calculating the volume of each slide and adding them up. This can be expressed as:

$$V_e = \sum_{i=0}^{n} \left[\frac{x_{i+1}-x_i}{2}\right]\left[\frac{h_i}{2}(w + L_i) + \frac{h_{i+1}}{2}(w + L_{i+1})\right] \quad (4.2)$$

Where,
x_i = distance from one end of embankment to chainage *i* (m); h_i = height of embankment from ground level at chainage *i,* (m); L_i = base width of dam at chainage *i,* (m); w = crest width of embankment (m); n = total number of chainages.

This can be written again as:

$$V_e = \sum_{i=0}^{n} \left[\frac{x_{i+1}-x_i}{2}\right]\left[\frac{h_i(2w+h_iS)}{2} + \frac{h_{i+1}(2w+h_{i+1}S)}{2}\right] \quad (4.3)$$

Where,
S = sum of the upstream and downstream slope. For example, if the upstream slope is 1:3 and the downstream slope is 1:2, then S = 3 + 2 = 5.

Estimating the reservoir capacity: The approximate formula for estimating the reservoir volume (Figure 4.3) is given as:

$$V_r = \frac{TLD}{6} \quad (4.4)$$

$$V_r = \frac{AD}{3} \quad (4.5)$$

Where,
V_r = volume of reservoir; T = maximum throw-back length of reservoir (m); L = maximum length of water in contact with dam at full supply level (m); D = maximum depth of reservoir near the dam (m); A = reservoir surface area at full supply level (m).

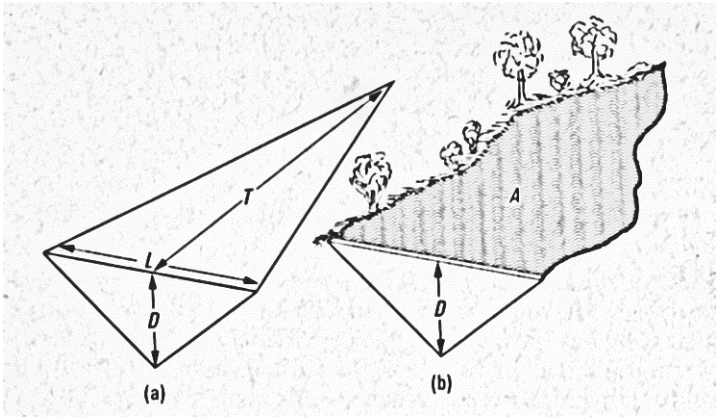


Figure 4.3 A typical reservoir behind a dam

A more accurate estimate can be obtained by the method of horizontal slices given as follows:

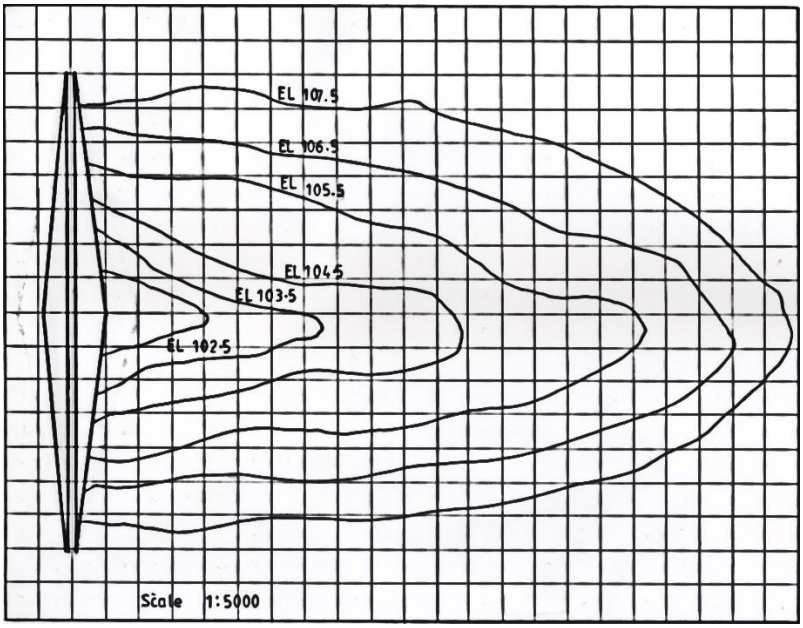


Figure 4.4 Method of horizontal slices for estimating reservoir volume

$$V_r = \sum_{i=1}^{n} \left[\frac{h_{i+1} - h_i}{2}\right] [A_i + A_{i+1}] \quad (4.6)$$

Where,
A_i = reservoir surface area at elevation *i* (m); h_i = elevation *i* (m); *n* = number of slices.

It is important to derive the stage-capacity and area-capacity curves for the reservoir during the design stage for the future management of the reservoir.

4.3 Design of Wells

Design of wells involve 1) hydrogeological investigations to determine the quantity, quality and yield of water through drilling and pumping test analysis; 2) installation of well casing and screens with gravel pack; and 3) installation of submersible pump for water delivery. Geological and hydrological data are required to produce the detailed design of a suitable well, the most important of which are:

- The thickness, character and sequence of strata to be drilled through.
- The properties (transmissivity and storativity) and the sequence of aquifers.
- The degree of confinement of the aquifer.
- The permeability of the confining strata.
- Water level and water level trends in the aquifer.
- The quality of water.
- The yield required from the well.
- The probable repercussions of the reduced water levels in the vicinity of the well.
- The case histories of any wells previously constructed in the area.

Occurrence of groundwater: Groundwater occurs in water bearing formations known as *aquifers*. Aquifers are not only porous but have high hydraulic conductivity or the ability to transmit water. There

are formations such as clays and shales that can store large amounts of water but transmit water very slowly and therefore referred to as *aquicludes*. Formations such as sandy clays are referred to as *aquitards* because they transmit water very slowly but which, over a large area of contact, may permit the passage of large amounts of water between adjacent aquifers which it separates. A rock which neither transmits nor stores water is called an *aquifuge*. An aquifer can be confined, unconfined, leaky and perched. *Confined aquifers* are sandwiched between two impermeable layers. Water in confined aquifers are usually under pressure. *Unconfined aquifers* are underlain by impermeable formations and exposed to the atmosphere. *Leaky aquifers* are bounded by aquitards and can be described as semi-confined since the overlying material is not entirely impermeable. While most aquifers are extensive in areal coverage, *perched aquifers* are not and unsustainable. Sedimentary rocks as sands, gravel, sandstones, limestone and chalk are typical aquifers, derived from fluvial, aeolian and glacial processes. Fissured igneous and metamorphic rocks can also be considered as aquifers.

Hydrogeological investigation: This involves examining hydrological, geological, hydrogeological and geophysical data available, with the intention of assessing any potential aquifers in an area. Relevant topographical and geological maps and possibly aerial photographs and satellite images should be studied to locate sites for possible wells. This should be followed by a reconnaissance survey, involving a walk over the site noting the soil and rock types present, the relief of the ground, the surface drainage and vegetation cover. Field exploration techniques used for the assessment of groundwater supply are largely governed by the local geological conditions. Geological mapping frequently forms the initial phase of exploration and these are done through drilling and studying the lithology of the area. A number of drilling methods exist, spanning the use of hand-held augers to the more sophisticated rotary-percussion drilling. The geological conditions influence the ease of drilling.

Geophysical methods are also used to determine the geological sequence and structure of subsurface rocks through the measurement of certain physical properties and forces. This procedure minimizes the number of drillholes in such investigations and hence reduce the cost of exploration. The geophysical methods include seismic, resistivity, magnetic and gravity methods. The seismic method involves the sudden release of energy from the detonation of an explosive charge in the ground, generating shock waves which radiate in hemi-spherical wave fronts from the point of release. The compressional and shear waves generate increase with depth below the surface and depend on porosity and mineralogy of the rock. The resistivity method applies in situations where aquifer thickness exceeds 6 m. Since most of the rock minerals are practically insulators, the resistivity of rocks and soils is determined by the amount of conducting mineral constituents and water content in the pores. Most soils and rocks conduct electric current only because they contain water. Rocks have magnetic properties induced by the earth's magnetic field. This property can be used to map geological structures. The earth's gravitational field varies according to the density of the subsurface rocks and may be used in reconnaissance geological surveys to reveal anomalies that may be subsequently investigated by other methods.

Well logging involves the plotting of the sequence of rocks in the subsurface. Logging of wells can be done using resistivity logging, induction logging, spontaneous potential logging, sonic logging, radioactive logging, temperature logging, fluid conductivity logging and caliper logging.

Pumping test analysis and determination of aquifer properties: Aquifers are porous and transmit water and the ability to transmit water determines the aquifer yield. As a porous material, the aquifer does not only yield water but also retain water. Its capacity to yield water is more important than the capacity to hold water. So, a certain proportion of the water in the aquifer can be removed by drainage under gravity,

while the remainder held by capillary forces. The ratio of the volume of water retained (V_{wr}) to that of the total volume of rock or soil (V) is referred to as the specific retention (S_{re}):

$$S_{re} = \frac{V_{wr}}{V} \tag{4.7}$$

The specific yield (S_y) of a rock or soil refers to its water yielding capacity attributable to gravity drainage (V_{wd}) as occurs when the water table declines and expressed as:

$$S_y = \frac{V_{wd}}{V} \tag{4.8}$$

The porosity (p) of the rock is the sum of the specific yield and specific retention as:

$$p = S_y + S_{re} \tag{4.9}$$

The most important aquifer properties are transmissivity (T) and storage coefficient (S). Pumping test is done on wells to determine these properties and consequently the yield of the aquifer. Pumping the well at a constant rate approximately equal to its design discharge allows the monitoring of well drawdown and recovery where the aquifer properties of T and S can be computed using the non-equilibrium equations. The Theim equilibrium equation, the Theis non-equilibrium equation, the Jacob non-equilibrium equation and the Chow non-equilibrium equation are used in the computation of aquifer transmissivity and storage coefficient.

The Theim equilibrium equation used for computing aquifer transmissivity is:

$$T = \frac{0.366Q}{d_w} log\left(\frac{R}{r_w}\right) \tag{4.10}$$

Where, d_w and $r_w$are well drawdown and radius respectively.
For this equation, T is insensitive to the radius of influence (R). This implies that a very large value of R is required for a small change in T. This method can be used for a good approximation of aquifer transmissivity.

The Theis non-equilibrium equation can be written as:

$$d = \frac{Q}{4\pi T}\left[-0.5772 - lnu + u - \frac{u^2}{2!2} + \frac{u^3}{3!3} - \frac{u^4}{4!4} + \cdots\right] \tag{4.11}$$

$$d = \frac{Q}{4\pi T} W(u) \tag{4.12}$$

Where,

$W(u)$ is the well function of u and d is the well drawdown.

$$u = \frac{r^2 S}{4Td} \tag{4.13}$$

By plotting two curves: 1) $W(u)$ against $1/u$; and 2) d against t/r^2 on logarithmic scales with the latter on a transparent paper and the two curves matched until they coincide. Any point under the matched

curves can be considered as a match point and the corresponding values of u, $W(u)$, d and t noted. These are substituted in equations 4.12 and 4.13 and the *T* and *S* computed as:

$$T = \frac{Q}{4\pi d} W(u) \tag{4.14}$$

$$S = \frac{2.25Tut}{r^2} \tag{4.15}$$

The Jacob non-equilibrium equation is given as:

$$d = \frac{2.303Q}{4\pi T} \log\left(\frac{2.25Tt}{r^2 S}\right) \tag{4.16}$$

This equation can be plotted as a straight line with d against $logt$.

For d = 0 and $t = t_o$,

$$\frac{2.25Tt_o}{r^2 S} = 1 \tag{4.17}$$

$$\frac{2.303Q}{4\pi T} = \Delta d \tag{4.18}$$

Thus, following from the equations 4.17 and 4.18, the properties of *S* and *T* can be computed as:

$$S = \frac{2.25Tt_o}{r^2} \tag{4.19}$$

$$T = \frac{2.303Q}{4\pi\Delta d} \tag{4.20}$$

The Chow non-equilibrium equation avoids the laborious curve fitting of the Theis non-equilibrium equation even though all assumptions and procedures are the same. By plotting the d-*log* t relationship as in the case of the Jacob equation, an arbitrary point is chosen on the curve and the coordinates of the point, d and t noted. The tangent to the resultant smooth curve is drawn through the chosen point and the drawdown per log cycle of time determined. The value of $F(u)$ is then computed as:

$$F(u) = \frac{d}{\Delta d} \tag{4.21}$$

Chow developed a nomogram relating $F(u), W(u)$ and u, thus enabling the use of equations 4.14 and 14.15 to compute the aquifer properties.

Well casing and screens: Well casing and screen serve the functions of supporting the side of the well from caving in and also keeping sediments out of the well. Commercial screens are available but can be made by creating slots in pipes of large diameters. The velocity of water as it enters through the screen slots is known as entrance velocity. Entrance velocities (v, m/s) of 0.03 m/s or less are preferred for optimum efficiency and can be computed as:

$$v = \frac{Q}{A} \tag{4.22}$$

Where,
Q = yield of well (m^3/s); A = open area of the screen (m^2).
The open area of the screen can be computed as:

$$A = \pi k D L \tag{4.23}$$

Where,
D= screen diameter (m); L= screen length (m); k= ratio of open area of screen to the total surface area.

Entrance velocities should not be allowed to exceed 0.045 m/s, else the screen slot area should be increased to reduce the entrance velocity. Screen slot sizes can be selected by first doing a particle size analysis on the aquifer material being screened. If the screen slot size is too large, the aquifer material may not stabilize and fine material will continuously be drawn into the well. If the screen slot size is too small, clogging may occur resulting in high friction losses and encrustation problems. Width of screen slot could be equal to particles sizes finer than between 40 and 60 %. Generally, 50 % finer than particle size (d_{50}) obtained from the particle size distribution curve could be used in the choice of screen slot width:

$$d_s = d_{50} \tag{4.24}$$

Where, fine material is drawn into the well, a sediment sump, which is 1 – 2 m length of casing, should be installed below the screen for the fine materials to settle without blocking the screen. Where gravel packs are required the gravel size can be computed as:

$$d_g \geq 5 d_s \tag{4.25}$$

Where,
d_g= effective gravel size (mm); d_s= screen slot size (mm)

Pump selection and installation: Submersible centrifugal pumps are preferred even though vertical shaft turbine pumps can be used. The submersible pump which can be used for lifts of up to 400 m, consists of a motor and the impeller coupled together. The pump unit requires pulling from the well when repairs or maintenance is required. The total head (H) for the pump can be computed as:

$$H = h_L + h_f + h_v + h_s \tag{4.26}$$

Where,
h_L= total vertical lift (m); h_f= friction headloss (m); h_v= velocity head (m); h_s= suction head (m).

4.4 Exercises

1. List the typical features of an earth dam, comprising the embankment and the reservoir.

2. Estimate the volume of earthworks for an earth dam with upstream and downstream slopes of 1: 3 and 1:2 respectively and crest width of 3 m. The maximum height of the dam is 4.5 m at the center of the 500 m long embankment. Using the method of vertical slices at 50 m intervals, estimate the volume of the embankment. Present your solution in a tabular form only.

3. Data on the reservoir capacity (V_r) of an irrigation dam are presented in the table below.

Elevation, h_i (m *asl*)	Reservoir surface area, A_i (ha)	Remarks
99.40	33.95	Ground level at reservoir bottom
100.00	34.48	
101.00	36.10	Dead storage level
102.00	38.00	
103.00	41.45	
104.00	45.04	
105.11	53.92	Full supply level

a) What is the reservoir capacity at full supply level?
b) Compute the live storage of the reservoir.
c) Compute the dead storage of the reservoir.
d) What is the maximum depth of the reservoir at full supply level?
e) If the water line extends 578 m on a 635 m long embankment, what should be the approximate throw-back length for this reservoir?

4. A well was drilled in a confined aquifer for irrigation purposes in the coastal savannah belt of Ghana. Calculate the pumping rate for 30 days of continuous pumping required to produce an easily measured drawdown of 3 m at an observation well 200 m from the pumping well. The transmissivity of the aquifer and the coefficient of storage are 2×10^{-2} m^2/s and 1.95×10^{-4}. Use the Jacob equation. If the water was pumped for irrigation for 2 hours on a 1.0 ha farm at the rate computed in the question above, what was the depth of water supplied by the well to the farm? If the irrigation efficiency was 80%, compute the average rooting depth of the crop irrigated for the following soil conditions: field capacity water content = 18.60 % mass; permanent wilting point water content = 7.83 % mass; specific gravity of dry soil = 1.40; allowable soil water depletion factor = 50%.

5. A 30 cm diameter well is to be developed for rural water supply and a pumping test was carried out. The well is pumped at a uniform rate of 4.50×10^{-2} m^3/s while observations of drawdown are made in a well 30 m away. Values of t and d are observed where t is the time after start of pumping in hours and d is the drawdown in meters. Assume the relation between r^2/t and d and between u and $W(u)$ are plotted and the match-point coordinates are:

$$u = 0.4;\ W(u) = 0.7;\ d = 1.04;\ r^2/t = 4.9 \times 10^3\ \text{m}^2/\text{d}$$

Further assume the following data and use linear interpolation where necessary:

u	1.0×10^{-3}	2.0×10^{-3}	3.0×10^{-3}	4.0×10^{-3}
$W(u)$	6.33	5.64	5.23	4.95

Compute the transmissivity, T, and the storage coefficient, S, of the aquifer. Also, estimate the drawdown, d, in the observation well at the end of 25 days of continuous pumping.

Module 5: Engineering Hydraulics

5.1 Introduction

This module will focus on hydraulics as applied to irrigation and drainage. Hydraulics is the study of the rest and movement of fluid substances such as liquids and gases. Hydraulics deals with liquids at rest (hydrostatics) and in motion (hydrodynamics). Subsequent sections will be directed at discussing the principles of closed and open channel hydraulics as applied to irrigation and drainage.

5.2 Pipe Hydraulics

The Reynolds number: Flow through pipes experience friction and shear tension. Friction occurs between the liquid and the wall of the conduit due to the ability of the liquid to cling or adhere to walls of the pipe. Friction therefore restricts movement of the water in the pipe. Water moves faster at the center of the pipe than close to the walls since the cohesive forces tend to be less than the adhesion near the pipe wall. The shear tension narrowly coincides with the viscosity and velocity of the liquid. Therefore when the effect of adhesion and cohesion are in equilibrium with the force exerted thereon, the average flow velocity will remain constant.

Pipe flows change from laminar to turbulent flow depending on the velocity of flow and these flow rregimes are described by the Reynolds number. The Reynolds number (R_e) is determined as:

$$R_e = \frac{d_i v \rho}{\mu} \tag{5.1}$$

Where,

ρ= density of water (kg/m^3)
μ = dynamic viscosity (kg/m.s)
v= average velocity of flow (m/s)
d_i= internal diameter of the pipe (m)

For laminar flow, $R_e \leq 2000$ and for turbulent flow, $R_e \geq 4000$. The flow is said to be in transition if the Reynolds number is between 2000 and 4000. Laminar flow occurs at low velocities and turbulent flow at high velocities.

Flow types: The continuity equation is given as:

$$Q = Av \tag{5.2}$$

Where,

Q = discharge (m^3/s)
A = cross-sectional area of flow (m^2)
v = velocity of flow (m/s)

Two main groups of flow occur: 1) steady and unsteady flow; and 2) Even and uneven flow. Steady flow occurs when the parameters Q and v do not vary with time, as measured at a specific section. Unsteady

flow occurs when these parameters vary with time at a specified section on the pipeline. Even flow occurs when the parameters Q and v do not change with distance. Uneven flow occurs when these parameters vary with distance.

Pressure distribution in pipelines: By application of Bernoulli's law, the total head available at any point in a pipeline is equal to pressure head plus elevation head plus velocity head. The difference in total head between any two points 1 and 2 on a pipeline under steady-state flow conditions is equal to the friction head loss between the two points:

$$\left(\frac{P}{\gamma}\right)_1 + z_1 + \left(\frac{v^2}{2g}\right)_1 = \left(\frac{P}{\gamma}\right)_2 + z_2 + \left(\frac{v^2}{2g}\right)_2 + h_f \quad (5.3)$$

The energy grade line describes the variation of total head along the pipeline, including velocity head, when flow occurs. The hydraulic grade line is defined by the energy grade line minus the velocity head. The hydraulic grade line defines the working pressure available at any point along the pipeline. The pressure distribution in a pipeline is determined by the relative positions of water along the line and friction head loss.

Pipe friction equations: These are used in the design of pumped irrigation systems. With designs where pipe friction losses are small, this can be ignored. Where, in totality, low pressure heads occur, the friction loss computations become important. There are a number of pipe friction head loss equations.

The Darcy-Weisbach equation is given as:

$$h_f = \frac{flv^2}{2gd_i} \quad (5.4)$$

Where,
h_f = friction headloss (m)
l = length of pipe (m)
d_i = internal diameter of pipe (m)
v = average flow velocity in pipe (m/s)
g = acceleration due to gravity (m/s^2)
f = friction factor

For laminar flow conditions,

$$f = \frac{64}{R_e} \quad (5.5)$$

For turbulent flow conditions,

$$\frac{1}{\sqrt{f}} = -2log\left(\frac{k}{3.7d_i} + \frac{2.51}{R_e\sqrt{f}}\right) \quad (5.6)$$

Where,
k = equivalent pipe roughness factor (mm)

Standard tables exist relating the *k* value to various pipe types.

The Hazen-Williams equation is given as:

$$h_f = l\left(\frac{3.59Q}{C_H d_i^{2.63}}\right)^{1.852} \quad (5.7)$$

Where,
h_f= friction headloss (m)
l = pipe length (m)
d_i= pipe internal diameter (mm)
Q = flow rate (m3/s)
C_H= Hazen-Williams coefficient

This equation is generally used in turbulent flow regime $(R_e > 4000)$ for pipes with diameters larger than 100 mm. Standard tables showing C_H values for pipe types are available.

There are other friction head loss equations such as Lamont, Blasius and Scobey equations.

Secondary pressure losses: Pressure losses occur on all installations on a pipe line. These could be valves, bends, Ts (Tees), elbows and reducers. In practice, these losses often make up such a small percentage of the total friction that they become negligible or are nominally allowed for as an estimated percentage of the pipe losses. In practice, these losses are usually expressed in terms of equivalent length of straight pipe of the relevant diameter or alternatively a velocity loss factor. Standard tables exist on pipe accessories and their equivalent losses.

Unsteady flows and water hammer: Pipes are designed to withstand certain minimum pressures depending on the pipe material. Water hammer is pressure surge in pipelines as a result of changes in water flow velocities. Changes in flow velocity cause the kinetic energy associated with velocity t be converted to pressure. The more rapid the change in flow velocity, the greater will be the magnitude of the resulting pressure surge. Pressure surges occur when valves are opened or closed, when pumps are started or stopped, or by sudden releases of entrapped air.

The magnitude of pressure surges associated with water hammer can be computed as:

$$\Delta H = \frac{a}{g}\Delta v \quad (5.8)$$

Where,

ΔH = surge pressure (m)
a = pressure wave velocity (m/s)
g = acceleration due to gravity (m/s^2)
Δv = change in water velocity (m/s)

The velocity or celerity of pressure wave is a function of both pipe material properties and fluid properties. The pipe material properties which affect the celerity are the modulus of elasticity of pipe material, Poisson ratio of pipe material, diameter and wall thickness of pipe. The fluid properties of major importance are the modulus of elasticity of water, fluid density and amount of air entrained in the fluid.

Pipe installations: Recommendations exist regarding pipe installations and standard tables are available on trenching requirements, specifying trench width and depth. Securing the pipe in position and prevent it from shifting in the trench is required to increase the capability of the line to resist deformation and failure at the joints. This is known as thrust blocking. Practically, thrust blocks are required where pipeline changes the direction of water flow, where pipe sizes change, at the end of a pipeline and where there is an in-line flow control valve. Standard recommendations exist on pipeline thrust loads and the bearing strength of soils.

Pumps: Pumps are used in irrigation and drainage. Different types of pumps are available and capable of doing specific jobs in water delivery. Pumps have certain performance characteristics and the most important being the head or pressure and discharge. Pressure and discharge are inversely related in pump design, meaning pumps which produce high pressure have relatively small discharge and vice versa. The specific speed (N_s) of the pump, which is closely related to the maximum operating efficiency of the pump can be computed as:

$$N_s = 0.2108N\left(\frac{Q^{0.5}}{H^{0.75}}\right) \tag{5.8}$$

Where,

N = rotational speed of pump (rpm)
Q = pump discharge (l/min)
H = pressure head (m)

The specific speed is a dimensionless factor.

The hydraulic principle for pumps relate volumetric flow rate (Q), pressure head (H), power rating (P), the impeller diameter (D) and the shaft rotational speed (N) as follows:

$$\frac{Q_1}{Q_2} = \frac{N_1}{N_2} \tag{5.9}$$

$$\frac{Q_1}{Q_2} = \frac{D_1}{D_2} \tag{5.10}$$

$$\frac{H_1}{H_2} = \left(\frac{N_1}{N_2}\right)^2 \tag{5.11}$$

$$\frac{H_1}{H_2} = \left(\frac{D_1}{D_2}\right)^2 \tag{5.12}$$

$$\frac{P_1}{P_2} = \left(\frac{N_1}{N_2}\right)^3 \tag{5.13}$$

$$\frac{P_1}{P_2} = \left(\frac{D_1}{D_2}\right)^3 \tag{5.14}$$

These are known as affinity laws. For a particular pump, manufacturers produce pump performance curves which graphically relate the head-discharge relation, pump efficiency and shaft input requirements of the pump.

Discharge and head requirements in irrigation will necessitate the combination of different pumps. If the aim is to have high discharge at low heads, pumps are installed in parallel. If the aim is to deliver at high heads for low discharge, pumps are installed in series.

Consider two pumps 1 and 2 with discharge Q_1 and Q_2, head H_1 and H_2 power rating P_1 and P_2. The following conditions apply if the pumps are installed in series:

$$Q_{series} = Q_1 = Q_2 \quad (5.15)$$
$$H_{series} = H_1 + H_2 \quad (5.16)$$
$$P_{series} = P_1 + P_2 \quad (5.17)$$

The following conditions apply if the pumps are installed in parallel:

$$H_{para} = H_1 = H_2 \quad (5.18)$$
$$Q_{para} = Q_1 + Q_2 \quad (5.19)$$
$$P_{para} = P_1 + P_2 \quad (5.20)$$

5.3 Open channel hydraulics

The Bernoulli law in equation 5.3 applies in open channel flow but the pressure head (P/γ) used in pipe hydraulics is replaced by the flow depth in channel. Flows can be steady and unsteady. Steady flows can be uniform and varied. For uniform flow, depth is the same at every section along the channel. Varied flow conditions can be gradual or rapid. For gradually varied flow, the change of depth of flow per unit length of channel is small. A rapidly varied flow is used to describe the flow which has abrupt changes in flow depth such as over a weir.

Reynolds number also applies in the description of open channel flow which is the ratio of the inertial forces to the viscous forces acting on the fluid. In laminar flow, the viscous forces predominate. In the turbulent flow, inertial forces predominate. For open cannel flow, Reynolds number (R_e) can be written as:

$$R_e = \frac{\rho L v}{\mu} \quad (5.21)$$

Where,

ρ = density of water (kg/m^3)
L= characteristic length (m)
v= flow velocity (m/s)
μ = dynamic viscosity (Ns/m^2)

But the kinematic viscosity (ϑ) is ratio between dynamic viscosity and fluid density:

$$\vartheta = \frac{\mu}{\rho} \quad (5.22)$$

For open channel flow, the characteristics length (L) is the hydraulic radius of the channel. In open channel flows, flow is said to be laminar if R_e is less than 500 and turbulent if greater than 4000.

Critical state of flow: Critical flow is the state of flow for which the specific energy is a minimum for a given discharge. The state of flow can be designated using the Froude Number (F_N) as:

$$F_N = \frac{v_{avg}}{\left[g\left(\frac{A}{T}\right)\right]^{1/2}} \quad (5.23)$$

Where,

v_{avg}= average flow velocity (m/s)
g = acceleration due to gravity (m/s^2)
A = cross-sectional area of flow (m^2)
T = top width of flow in channel (m)

The following states f flow can therefore be designated with respect to the Froude Number:

- F_N= 1 represents critical flow.
- $F_N > 1$ represents supercritical flow
- $F_N < 1$ represents subcritical flow

Channel design for uniform flow: Open channels are designed using the Manning's equation which is given as follows:

$$Q = \frac{A}{n}(R)^{2/3}S^{1/2} \quad (5.24)$$

Where,

Q = discharge (m^3/s)
A = cross sectional area of flow (m^2)
R = flow hydraulic radius (m) = $\frac{A}{P}$
P = wetted perimeter of flow (m)
S = channel bed slope
n = Manning's roughness coefficient

Channels for carrying flow can have different cross sections. Rectangular, trapezoidal, triangular and circular sections are among the commonly used design sections. Channels can be lined or unlined. The channels must be such that flow velocities are non-erodible, whether lined or unlined. For lined channels, unless otherwise specified, maximum permissible flow velocity used as guide is 1.5 m/s. The channel must have the best hydraulic section. That is, the channel should have the least wetted perimeter for a given cross-sectional area of flow, constituting the most hydraulically efficient section in transporting water. This is also referred to as economic channel section.

For a hydraulically efficient trapezoidal section, the following condition holds:

$$b = 2d\left[(N^2 + 1)^{1/2} - N\right] \quad (5.25)$$

Where,

b = channel bed width (m)
d = depth of flow in channel (m)
N = channel side slope

For a hydraulically efficient rectangular section, *N* = 0, so that:

$$b = 2d \tag{5.26}$$

Other relationships exist for different channel sections and can be found in standard hydraulics textbooks. The principle of basing the design on the best hydraulic section applies only to non-erodible channels. Designs for channels constructed of erodible materials must consider maximum recommended flow velocities and the distribution of forces necessary to maintain channel stability.

Erodible channels are designed based on the critical tractive force theory and he maximum velocity criterion. These are channels cut through the earth material. The stability of the channel material and its ability to resist particle detachment and transport by moving water forms the basis for this design. There are recommended non-erodible flow velocities for different earth materials. The critical tractive force theory states that the tractive force of the channel material should be kept below a critical value so as not cause channel erosion. For a trapezoidal section,

$$\tau_{cs} = \tau_{cb}\sqrt{1 - \frac{\sin^2\theta}{\sin^2\varphi}} \tag{5.27}$$

$$\tau_{cs} = 0.76\rho g d S \tag{5.28}$$

Where,

τ_{cs}= critical tractive force on the side of trapezoidal channel (N/m^2)
τ_{cb}= critical tractive force on bed of trapezoidal channel (N/m^2)
θ = angle of inclination of the side of channel to horizontal
φ = angle internal soil friction
ρ= density of water (kg/m^3)
g = acceleration due to gravity (m/s^2)
d = depth of flow in channel (m)
S = bed slope of channel (m/m)

5.4 Exercises

1. Fill in the blank spaces below

   a) Engineering hydraulics involves the study of _____.
   b) In surface tension, the liquid surface behaves as if it were an _____ membrane under tension.
   c) The Hazen-Williams equation is used for computing_____.
   d) The discharge of a pump is proportional to the _____ of the impeller diameter of the pump.
   e) Water hammer is caused by rapid changes in _____ in the pipeline
   f) _____ prevents the pipeline from shifting in the trench and is required to increase the capability of the line to resist deformation and failure at the joints.
   g) For an economic, trapezoidal channel (side slope = 1/N), a general relationship between the bed width (b) and depth (d) of flow is given as_____.

h) For the same condition in *question 1 (g)*, the cross sectional area (*A*) of flow in the channel for N = 1 is expressed in terms of the flow depth as_____

i) For the same condition in *question 1 (g)*, the wetted perimeter (*P*) of flow in the channel for N = 1 is expressed in terms of the flow depth as_____

j) For the same condition in *question 1 (g)*, the hydraulic radius (*R*) of flow in the channel for N = 1 is expressed in terms of the flow depth as_____

2. Design a lined rectangular open drain to carry 0.93 m^3/s for the following conditions: Bed width = 1.4 m; Canal bed slope = 1:500; Manning's roughness coefficient = 0.015. Assume a freeboard of 20%. Compute: 1) the depth of flow and 2) the velocity of flow.

3. You are required to design an *unlined trapezoidal* irrigation canal to be excavated in silty clay loam using the following design data:

Critical tractive force on channel bed	=	2.4 N/m^2
Angle of internal soil friction	=	30°
Canal discharge	=	10 m^3/s
Canal bed slope	=	1/10,000
Manning's roughness coefficient	=	0.02
Side slope of canal	=	½
Freeboard	=	20 % of design depth

Check that the flow velocity is *non-erodible* if the maximum permissible velocity for this soil is 1.0 m/s.

4. An existing irrigation system has a pump which delivers 3,000 l/min flow at a total head of 62 m. The impeller diameter is 25 cm and it is rotated at 1750 rpm. A motor with an output shaft power of 40.5 kW is required to drive the pump. The existing irrigation system is modified to incorporate low pressure nozzles and make other reductions in friction head loss to 50 m head. It is decided to keep the existing pump but to pull and trim the impeller to match the new system requirements.
   a) Determine the impeller diameter required for the new discharge conditions and the power necessary to drive the modified pump.
   b) Determine the flow rate which will have to be produced by an additional pump to meet the original discharge requirements when the modified pump is re-installed.

5. A pipeline with 200 mm inside diameter and 340 m length made of new PVC is laid along a horizontal grade. The required flow rate in the pipeline at steady state is 40.0 l/s and the total head available at the inlet is 330 kPa. Compute the working pressure head at the distance 340 m from the inlet by application of the Darcy-Weisbach headloss equation. Flow is fully turbulent. Assume the conversion factor, k = 8.263 x 10^7; $\varepsilon = 0.015$ mm.

$$h_f = kfL\frac{Q^2}{D^5}$$

$$\frac{1}{\sqrt{f}} = 2\log\left(3.7\frac{D}{\varepsilon}\right)$$

where, h_f = friction headloss (m)
f= Darcy-Weisbach factor

L = length of pipe
Q = discharge (l/s)
k = conversion factor
D = pipe diameter (mm)

Module 6: Irrigation System Design

6.1 Introduction

In addition to the development of water resources for irrigation as explained in module 4, systems layout design is required for water conveyance, distribution and application at the field level. Contour map for the area will be required and since water delivery is by gravity, field grading will be required. The irrigation systems to be considered here include:

- Surface irrigation (furrow, border and basin)
- Sprinkler irrigation
- Drip or trickle irrigation

The considerations for the design of these irrigation systems are explained in the subsequent sections.

6.2 Surface irrigation design

Surface irrigation requires flooding the soil surface at the head of the field and as the flow advances by gravity on the surface, it infiltrates into the soil at the same time. The surface irrigation include furrow, basin and border irrigation systems. Simple as surface flooding may seem, achieving a good water application efficiency can be daunting. It depends on:

- Irrigation stream size
- Slope
- Advance
- Recession
- Infiltration

Since water application is by surface flooding, the objective is to minimize the runoff and percolation losses to improve field efficiency. For sandy soils (light soils), stream sizes must be relatively large and field lengths relatively short to minimize losses. Since the basic infiltration rates are high for light soils, the intention is to quicken the advance process at the same time being mindful of the high rates of infiltration. For clay soils, stream sizes must be relatively small with relatively long field lengths to minimize field losses. The basic infiltration rates are lower and hence requires a much longer contact time for the water to infiltrate into the soil while being mindful of the runoff losses.

Irrigation stream size: Irrigation stream size (q, l/s) is given as:

$$q = \frac{27.78(\theta_{FC}-\theta_{PWP})p\rho_d ZA}{E_a t} \quad (6.1)$$

Where,

θ_{FC}= field capacity water content (% mass)
θ_{PWP}= permanent wilting point water content (% mass)
p = allowable soil water depletion factor
ρ_d= specific density of dry soil
Z = average rooting depth (m)

A = farm size (ha) = inter-row spacing x field length
t = irrigation time (h)
E_a= field water application efficiency

Contact time: Since surface irrigation requires flooding the surface and letting the water run the length of the field, it is obvious that some parts will receive water for a longer period than others. By inference, the head of the field where the water is applied first will receive more water than the tail end of the field. The time the water is in contact with the soil is known as contact time and it is equivalent to the difference between recession and advance times.

$$t_c = t_r - t_a \tag{6.2}$$

Where,
t_c = contact time
t_r = recession time
t_a = advance time.

Advance, recession and infiltration curves: The contact time and average depth of water infiltrated along the field length are determined from the advance and recession curves and also the infiltration curve as follows:

- The rate of movement from the end of the field where is applied is monitored with time and the advance curve of time (*t*) against field distance (*x*) is plotted.
- The irrigation stream is cut off at the head of the field and the rate of water recession along the field length (x) is monitored with time (*t*) since the time of cutoff . This is also plotted on same graph.
- The contact time (t_c) can be determined from the graph from the beginning of the field to the end at regular distances.
- Using the double-ring infiltrometer test, infiltration data can be obtained and plotted.
- Two curves can be plotted on the same diagram. The cumulative infiltration (*D*, mm) - time (*t*, min) curve and infiltration (*i*, mm/h) – time (*t*, min) curve. The infiltration rate curve fits the Kostiakov equation as:

$$i = kt^n \tag{6.3}$$

- By integrating this equation, the cumulative infiltration curve can be obtained as:

$$D = ct^m \tag{6.4}$$

Where,

$$c = \frac{k}{n+1} \tag{6.5}$$

$$m = n + 1 \tag{6.6}$$

Determination of appropriate field length: Appropriate field length in surface irrigation, given soil characteristics, slope and stream size are determined based on the *T/4* rule, where *T* is the time required

for infiltrating the cumulative amount of irrigation into the soil. Using the advance curve, the corresponding field length to *T/4* is the appropriate field length required to minimize field losses.

Determination of field water application efficiency (E_a): The following procedure can be followed in the determination of field water application efficiency:

- By determining the contact times along the irrigation field length, the amounts of water infiltrated into the soil along the field length can be determined from the cumulative infiltration–time curve.
- If the depths of water infiltrated at distances $x_1, x_2, x_3, \ldots\ldots, x_n$ are $D_1, D_2, D_3, \ldots\ldots\ldots, D_n$ respectively, then the average depth of infiltration ($\overline{D}$) along the field and the field water application efficiency are given as:

$$\overline{D} = \frac{\sum_{i=0}^{n} D_i}{n} \tag{6.7}$$

$$E_a = \frac{10(\theta_{fc} - \theta_{pwp}) p \rho_d Z}{\overline{D}} \tag{6.8}$$

6.3 Sprinkler irrigation design

Sprinkler irrigation is intended to mimic rainfall. This does not require the grading of the land. Several sprinkler types were developed in response to several needs. These include hand-move, side-roll, center pivot and the rain gun. The typical system includes a pump, a mainline, a lateral and the sprinkler nozzles. The laterals and the sprinklers are spaced appropriately to meet the cropping density. Design considerations include sprinkler uniformity as affected by the nozzle pressure and wind and adequacy of water application. Sprinkler manufacturers will normally supply these specifications for the hand-move systems especially:

- Nozzle operating pressure
- Nozzle discharge
- Sprinkler x lateral spacing
- Sprinkling rate
- Radius of throw

Sprinkler application is designed to overlap each other to achieve some sprinkling uniformity. In a calm wind situation, the sprinkling distribution appears normally distributed. In the event of a strong wind effect, the sprinkling distribution tends to be skewed in the direction of the wind. Similarly, adequate nozzle operating pressure results in a normally distributed sprinkling pattern. Low and high pressures negatively affect the distribution pattern and hence sprinkling uniformity. Sprinkler discharge ($q, l/s$) is function of nozzle operating pressure (p, kPa):

$$q = Kp^{0.5} \tag{6.9}$$

Where,

K = nonlinear proportionality constant dependent of model and diameter of nozzle.

Sprinkler system capacity: Sprinkler system capacity (Q) is computed using the following equation

$$Q = 2.778 \frac{I_G A}{N_{op} T_{op}} \quad (6.10)$$

Where,
Q = continuous flow rate required (*L/s*); I_G = gross irrigation requirement (mm); A = total irrigated area (ha); N_{op} = number of days of operation per irrigation interval (d); T_{op} = hours of operation per day (h/d)

Nozzle sprinkling rate: This can be computed as follows:

$$i_s = \frac{q}{L_L x L_S} \quad (6.11)$$

Where,
i_s = sprinkling rate (mm/h); q = nozzle discharge (L/h); L_L= lateral spacing (m); L_S= sprinkler spacing (m)

For the selection of the sprinkler type, the sprinkling rate should be less than or equal to the soil infiltration rate. This is intended to minimize runoff losses.

Lateral discharge: This can be computed as:

$$Q_l = Nq \quad (6.12)$$

Where,
Q_l= lateral discharge (L/h); q= sprinkler or nozzle discharge (L/h); N= number of sprinklers on the lateral

Hydraulics of Laterals: The design of laterals is based on the following assumptions: 1) Sprinklers are evenly spaced; 2) Discharge is the same for each nozzle; 3) Total flow into the laterals is discharged through the nozzles. The actual headloss in the lateral due to friction can be computed as:

$$H_{L-ac} = F H_{L-p} \quad (6.13)$$

Where,
H_{L-ac}= actual head loss due to friction (m/m); F = friction factor to account for decrease in flow along lateral; H_{L-p}= equivalent head loss due to friction in a through-flow pipe (m/m).

The procedure for lateral design requires a balance to be developed between the length of lateral, the headloss due to friction in the lateral and the change in elevation head due to topographic effects. These factors are kept in balance so that the pressure variation between two most critical sprinklers on the lateral is limited to $\pm$ 20 %. The governing equation for the maximum allowable head loss due to friction between two critical sprinklers is given as:

$$H_L = \frac{[\theta H_a - H_e]}{L} \quad (6.14)$$

Where, H_L= maximum allowable headloss due to friction (m/m); H_a= nozzle operating pressure (m); H_e= difference in elevation head between two most critical sprinklers (m); L = length of lateral between two critical sprinklers (m)

Laterals are designed based on friction headloss formulae such as the Hazen-Williams equation, Darcy-Weisbach equation and Scobey equation. There are other friction headloss formulae in addition to these. Laterals are assumed first as through-flow pipes and the theoretical headloss computed. The friction factor is applied to compute actual headloss as shown in equation 6.13.

Hazen-Williams friction head loss formula for a through-flow pipe can be written as:

$$h_f = kL\left[\frac{(\frac{Q}{C})^{1.852}}{D^{4.87}}\right] \tag{6.15}$$

$$H_{L-p} = \frac{h_f}{L} \tag{6.16}$$

The *F*- factor accounting for the decreasing flow rate along the lateral are for two sprinkler arrangements. When the first sprinkler on the lateral from the mainline is at the same distance as the spacing between subsequent sprinklers, the *F*-factor can be written as:

$$F = \frac{1}{m+1} + \frac{1}{2N} + \frac{(m-1)^{0.5}}{6N^2} \tag{6.17}$$

When the first sprinkler on the lateral from the mainline is at half the distance as the spacing between subsequent sprinklers, the *F*-factor can be written as:

$$F = \frac{2N}{2N-1}\left(\frac{1}{m+1} + \frac{(m-1)^{0.5}}{6N^2}\right) \tag{6.18}$$

Where, m = exponent on velocity head in friction headloss formula; N = number of nozzles on a lateral.

Mainline system design: The pressure requirement at the mainline entrance to the lateral can be computed as:

$$H_n = H_a + 0.75\left(H_f + H_e\right) + H_r \tag{6.19}$$

Where, H_n= pressure required at mainline entrance to lateral (m); H_a= nozzle operating pressure (m); H_f= friction headloss between two most critical sprinklers (m); H_e= difference in elevation head between two most critical sprinklers (m); H_r= height of riser (m). The factor 0.75 is to produce the average operating pressure near the mid-point of the lateral.

The pressure required at any point of interest on the mainline is computed as:

$$H_i = H_n + h_{f-in} + H_{e-in} + H_{v-in} \tag{6.20}$$

Where, H_i= pressure required at point *i* on mainline (m); H_n= pressure required at point *n* on mainline (m); h_{f-in}= friction headloss between point *i* and point *n* in the direction of flow (m); H_{e-in}= elevation

head difference between point *i* and point *n* in the direction of flow (m); H_{v-in}= velocity head between point *i* and point *n* in the direction of flow (m).

The total dynamic head for the pump can be computed as:

$$TDH_i = H_i + h_{f-Pi} + H_{e-si} + h_{f-s} + \frac{v_i^2}{2g} \quad (6.21)$$

Where, TDH_i= total dynamic head at point *i* on the mainline (m); H_i= pressure required at point *i* on the mainline (m); h_{f-Pi}= friction headloss between the pump and point *i* (m); H_{e-si}= elevation head difference between level of water at source and point *i* (m); h_{f-s}= friction headloss in the suction pipe (m); $\frac{v_i^2}{2g}$= velocity head at point *i* on the mainline (m).

The maximum value for the total dynamic head computed is selected as the pump requirement.

Sprinkler Uniformity: Whenever a new sprinkler system is installed, it is run for some time to check any system leakages and sprinkler uniformity. Water collecting cans are set up in upright positions in a grid form within a grid of four sprinklers and operated at the required operating pressure for at least two hours. The water sprinkled by the four overlapping sprinklers are measured in each can and recorded. The sprinkler uniformity is then computed using Christiansen Uniformity Coefficient (Cu):

$$C_u = 100\left(1 - \frac{\sum_{i=1}^{n}|x_i - \bar{x}|}{\sum_{i=1}^{n} x_i}\right) \quad (6.22)$$

C_u = coefficient of uniformity; x_i = depth of water collected in the measuring can (mm); $\bar{x}$ = mean depth of water collected in the measuring can (mm). Coefficient of uniformity values of 75 % or more are deemed to be satisfactory. Else, the system operating pressure or the wind effect or a combination of the two could be the reasons for poor sprinkler uniformity.

6.4 Drip irrigation system design

A pressurized irrigation system, the design procedure is not too different from sprinkler system design. The conventional drip system comprises: 1) mainline: pump, primary filter, pressure gauge, discharge control valve, flow meter; 2) sub-main: secondary filter, solenoid valve for automation, secondary pressure gauges, flush valves; 3) lateral: lateral pipes, emitters. System performance characteristics include the system operating pressure, emitter flow rate, emitter uniformity, filtration and water treatment and chemical injection.

The system requires the laying of lateral pipes to every plant in the field and also lends itself to the combined process of irrigation and chemical application in one process, otherwise known as chemigation. Considered to be the most efficient of the irrigation systems, it is also the most expensive. Drip irrigation has high crop water productivity, that is, more yield per unit volume of water. Drip irrigation is also known as localised irrigation because the entire soil surface is not wetted but only the region near the crop. That means the soil does not have enough water buffer in case of equipment malfunction and this can lead to crop failure.

Types of emitters: These are at the delivery end of the water to the crop and their performance can have a profound effect on the development of the crop. There are different emitter designs. Some drip water at specific points and are referred to as point sources. There are others that behave like small leaky pipes that emit water along their entire length and therefore referred to as line emitters. Conventional emitter designs include (but not limited to) the single-exit long path emitter, single-exit orifice emitter, orifice-vortex emitter and the multiple-exit long path emitter. Flow through the emitter can be laminar or turbulent and can be categorized using the equation for computing the Reynolds number (Equation 5.1). The following categories of flow regimes can be categorized using the Reynolds number:

- $R_e \leq 2000$ Laminar flow
- $2000 < R_e \leq 4000$ Unstable
- $4000 < R_e \leq 10000$ Partially turbulent
- $R_e > 10000$ Fully turbulent

Number of emitters required: This can be estimated based on the farm area, the planting density and the number of emitters required per plant. Depending on emitter flow rate one can have more emitters per location to provide the required amount. This can be computed as:

$$N_e = AN_pN_{ep} \tag{6.23}$$

Where, N_e= number of emitters; A = area of farm (ha); N_p= planting density (number of plants per ha); N_{ep}= number of emitters per plant.

Required emitter discharge: The required emitter discharge can be computed based on the crop water requirements per day, distribution patter efficiency and system operating time in hours per day.

$$q_e = \frac{(10^4)AET_c}{N_eE_dt} \tag{6.24}$$

Where, q_e= required emitter discharge (l/h); ET_c= peak crop evapotranspiration (mm/d); E_d= distribution pattern efficiency (factor); t = system operating time (h/d)

Emitter discharge can also be computed as a function of the operating pressure.

$$q_e = Kp^n \tag{6.25}$$

Where, K, n = proportionality constants; p = emitter operating pressure (kPa).

Different equations relating emitter discharge and operating pressure are required depending on type of emitter and flow regime. The discharge equation for an orifice emitter in fully turbulent flow is given as:

$$q_e = 3.6aC_o\sqrt{2gH} \tag{6.26}$$

Where, a = emitter cross sectional area (mm^2); C_o = orifice coefficient; g= acceleration due to gravity (m/s^2); H= emitter operating pressure head (m).

The orifice coefficient is typically assumed to be 0.8 for the condition stated above. The governing equation for a long-path emitter for a laminar flow regime is given as:

$$q_e = 0.11384a\sqrt{\left[2g\left(\frac{HD}{fL}\right)\right]} \quad (6.27)$$

$$f = \frac{64}{R_e} \quad (6.28)$$

Where, D= lateral diameter (m); f= friction factor; L= emitter flow path length (m)

Emitter discharge for turbulent flow through a long-path emitter is given as:

$$q_e = 0.11384a\sqrt{\left[2g\left(\frac{D\sqrt{H}}{fL}\right)\right]} \quad (6.29)$$

$$\frac{1}{\sqrt{f}} = 2log\left(\frac{D}{\epsilon}\right) + 1.14 \quad (6.30)$$

Where, ϵ = absolute roughness of pipe or tubing (mm).

Emitter uniformity: This is to enable the uniformity of emitter discharge along the lateral to be assessed using: 1) emission uniformity; and 2) emitter flow variation. The emission uniformity is given as:

$$U_e = 100\left[1 - \left(\frac{1.27C_v}{n}\right)\right]\left(\frac{q_{min}}{q_{avg}}\right) \quad (6.31)$$

Where, U_e = emission uniformity (%); n (for point source) = number of emitters per plant for point source emitters on a permanent crop; n (for line source) = 1 or any higher value based on the ratio of plant spacing and emitter spacing on lateral line as specified by manufacturer; = manufacturer's specified coefficient of variation; q_{min}, q_{avg}= minimum or design emitter discharge respectively (l/h).
Emission uniformity of more than 70 % is considered satisfactory.

The emitter flow variation can be computed as:

$$q_{var} = 100\left(1 - \frac{q_{min}}{q_{max}}\right) \quad (6.32)$$

Where, q_{var}= emitter flow variation (%); q_{min}, q_{max}= minimum and maximum emitter discharge respectively (l/h). The emitter flow variation of 20 % or less is acceptable but 10 % or less is desirable. Else, it is not acceptable.

Lateral hydraulics: The hydraulics of laterals for drip irrigation is the same as for other distribution systems. The head losses in lines and across control devices are computed using the same procedures used in sprinkler irrigation system design already explained above.

Filtration and water treatment systems: Filters are used in drip systems to preclude any suspended materials in the irrigation water from clogging the emitter. Materials that can clog emitters can be suspended solids, chemical precipitates and biological materials produced by bacteria and algae. Dealing with the clogging problem can be achieved through back-flushing of filters, bacterial treatment to remove bacterial slime and chemical treatment to reduce the potential of precipitation of insoluble salts.

Chemical injection systems: Drip irrigation is the only irrigation system that lends itself to the combined process of irrigation and chemical application in one process, known as chemigation. Chemical injectors are installed on the sub-main. The most common type of injection system is an auxiliary positive displacement pump which feeds the chemical solution into the distribution system. The flow rate through the pressurized chemical holding tank is controlled by valves. The function for the concentration of material remaining in the tank as a ratio of the original concentration is given as:

$$C = 100\, e^{\left(\frac{-q_t}{100}(t)\right)} \tag{6.32}$$

Where, C = concentration of materials remaining in the tank as a ratio of the original concentration; q_t= flow rate through the tank per unit time as a ratio of tank capacity (%); t = time with units corresponding to q_t.

Flow rates through the tank can be controlled by valves and meters. Disk orifices installed in the tank discharge line can also control flow. The governing equation for the orifice diameter is given by:

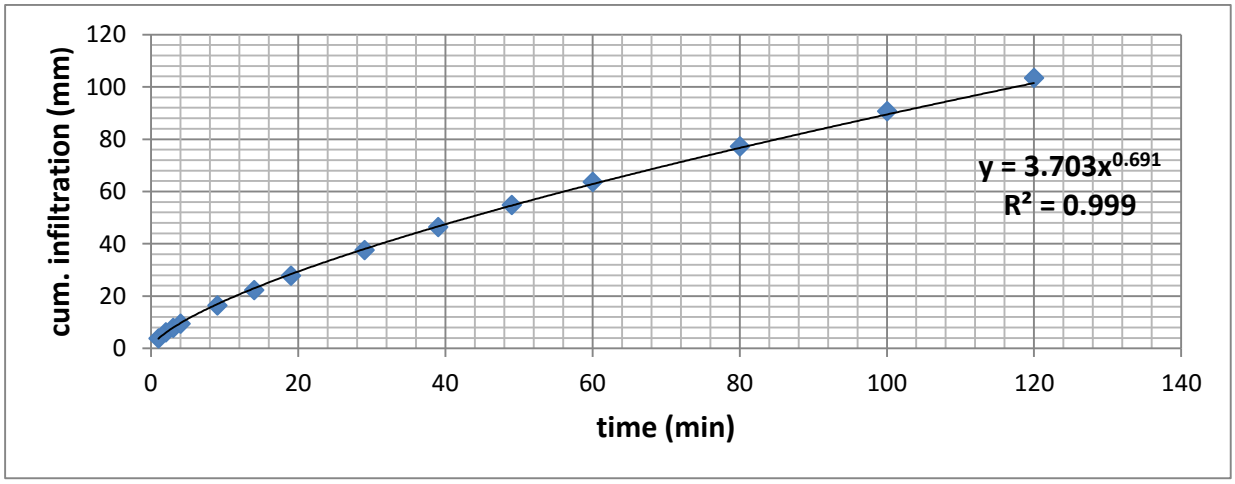
$$D = \sqrt{\frac{15.13 Q_t}{C_o \sqrt{P}}} \tag{6.33}$$

Where, D = orifice diameter (mm); Q_t= flow rate through the tank (l/min); C_o= orifice coefficient; P = pressure differential across the orifice (kPa). The standard value for the orifice coefficient is 0.62.

6.5 Exercises

1. Explain, *briefly in a logical sequence*, what you would do in surface irrigation design to select the right length of furrow that will minimize runoff and deep percolation losses?

2. The diagram below shows the cumulative infiltration (*D, mm*) against time (*t, min*) for a soil in an irrigation farm with the following characteristics: field capacity water content = 18 % mass; permanent wilting point water content = 6 % mass; soil water depletion factor = 0.5; soil dry specific gravity = 1.25; average soil rooting depth = 1.0 m.



What should be the approximate time of irrigation for the field conditions?

3. Assuming the terminal infiltration rate for the soil in *question 2* could be obtained 3 hours since infiltration started, what could be the infiltration rate of the soil? [Show the procedure for obtaining the answer]
4. Calculate the *maximum 24-hr* stream flow (m^3/s) for a proposed 100 ha sugarcane farm using the following data. The sugarcane is planted in September and takes 13 months to mature for harvesting. Assume a furrow irrigation efficiency of 70% and system flexibility factor of 1.1. The calculation must be clearly presented in a table on monthly basis.

Growth period (month)	1	2	3	4	5	6	7	8	9	10	11	12	13
Crop coefficient	0.3	0.5	0.7	0.9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.9	0.7

Month	J	F	M	A	M	J	J	A	S	O	N	D
ETo (mm)	261	235	185	169	210	185	210	187	177	210	220	220
Pe (mm)						30	136	162	50			

5. A trial configuration of a hand-move sprinkler system has a lateral running downslope from a mainline along a constant grade of 0.005 m/m. The design operating pressure of the nozzle is 310 kPa. The trial length of the lateral results in a distance of 400 m between the first and the last sprinkler. Compute the maximum allowable head loss due to friction.

6. A sprinkler nozzle discharges at 0.36 l/s. If the lateral x sprinkler spacing is 18 m x 18 m, what is the sprinkling rate of the sprinkler? Would you consider this nozzle suitable for the soil with an infiltration rate of 10 mm/h? (Yes or No?). [Show the procedure for obtaining the answer].

7. A distribution system is to be designed for a sprinkler system on which 5 laterals will be operated simultaneously. Each lateral is to have 30 nozzles with 0.360 l/s discharge per nozzle. The required pressure on the mainline at the inlet to each lateral is 393 kPa. The elevation of each point on the line and distance between points are given in the following table:

Point	Elevation (m)	Distance between points (m)
Pump	48.41	
		36.58
A	47.55	
		18.29
B	48.46	
		18.29
C	48.77	
		18.29
D	47.85	
		18.29
E	47.24	

8. The main is made of 20.32 cm inside diameter aluminum pipe with a Hazen-Williams C factor of 144. The friction loss and elevation head from the water surface to the centerline of the pump is +0.333m. Laterals are to operate on points A through E simultaneously. Determine the critical pressure point in the mainline and the total dynamic head requirement of the pump.

9. A typical orchard is to be developed on a field with dimensions 253 m by 439 m. The orchard will be irrigated using trickle system laid out so that each tree is served by four emitters. The following design conditions are based on peak period requirements at full tree maturity: operating pressure head at the emitter = 10 m; peak period crop water requirement = 5 mm/d; distribution pattern efficiency = 92 %; operating time = 18 h/d. The total lateral length should not exceed 1900m per ha of the field. Assume planting density of 250 trees per ha and 4 emitters per tree. Compute the number of emitters to the nearest whole number, the required emitter discharge and the required lateral length for the orchard.

10. After operating a drip irrigation system for 18 hours, the maximum and minimum flow rates on the drip lines were 5.7 and 4.3 l/h. Compute the coefficient of variation for the drip system. Will you consider the performance of the drip system in the above question ***desirable***, ***acceptable*** or ***not acceptable***?

11. A fertilizer injection tank with a capacity 350 l has a 2 mm diameter disk orifice installed on the discharge side. Under operating conditions, the pressure differential across the orifice is maintained at 20 kPa. Compute the discharge through the tank. Compute also the time required for the concentration of fertilizer in the tank to be 50 % of the original concentration.

Module 7: Agricultural Land Drainage and Reclamation

7.1 **Introduction**

Most agricultural crops require a well aerated root zone free of saturation water. Prime agricultural land are either naturally well drained or has some form of artificial drainage been introduced into the land to control both water table and excess surface water. Poorly drained soils are less desirable for efficient agricultural production. It has been found that in many cases good drainage makes the difference between a highly productive land and a marginal one. Lands that have salt problems tend to inhibit crop growth, requiring that such salt levels must be reduced to tolerable levels to enable crops to yield appreciably.

Agricultural land drainage and reclamation involves:

- Controlling excess water on the soil surface;
- Controlling excess water in the soil; and
- Controlling excess salts within the soil, if any.

7.2 **Causes of drainage and salt accumulation problems**

The causes of excess water and salts in the agricultural land can be attributed to the following:

High rainfall: Areas with substantially high rainfall are prone to poor drainage leading to excess water on and underneath the soil surface.
Artesian flow from deep aquifers: This can result in high water tables resulting in soil submergence.
Overland flow in low lying areas: Low lying areas naturally receive runoff from surrounding uplands resulting in land submergence.
Poor water management in irrigated areas: Once water is continually being applied to the soil especially in case of over-irrigation, there is the possibility of raising the water table if the soil is poorly drained. Sea water intrusion in coastal areas: Coastal areas are prone to sea water intrusion and where such lands are below sea level, thus resulting in high water tables and salt problems in such coastal lands.
Shallow hard pans: Some lands have shallow concretions at shallow depth and cause rainfall building up to the land surface since percolation is inhibited.
Land underlain by salty rocks: Lands underlain by salty rocks tend to cause salt accumulation in such lands.
Agrochemical accumulation and high evaporation: Due to intensive farming in irrigated areas, the continuous application of agrochemicals can accumulate in the soil and adversely affect the crops if measures are not taken to leach the salts and drain the poor quality water out of the irrigated field.

7.3 **Adverse effects of poor drainage**

Poor drainage and excess salts affect the productivity of lands due to the following reasons:

- There is lack of soil aeration thus affecting the activity of soil bacteria and retarding the formation of organic acids needed to dissolve plant food materials.
- Flow of water from salt laden strata underneath results in the accumulation of salts in the root zone of the crops. Crops are destroyed or yields drastically reduced.
- There is difficulty in carrying out cultivation operations in poorly drained land.

- Nitrification process is reduced due to the presence of excess water reduces the supply of oxygen.
- Under water-logging conditions, soil temperature drops and consequently affect seed germination.
- Crop maturity periods are adversely affected in poorly drained lands.
- In poor drainage conditions, soil chemical reactions are affected and consequently reducing the uptake of soil micronutrients by the crops.
- Poor drainage increases soil pH resulting in yield reductions and even soil infertility.

7.4 **Benefits of agricultural land drainage**

Agricultural land drainage benefits can be related to: 1) crop growth; 2) livestock production and 3) soil cultivation.

Drainage benefits related to crop growth: Crops benefit from good drainage as follows:

- Increased soil volume available for plant roots;
- Plant roots have access to more nutrients;
- A well-drained soil warms up more quickly; and
- More activity by microbes, earthworms and other beneficial organisms in the soil when soil is well-drained.

Drainage benefits related to livestock production: Livestock benefits from good drainage as follows:

- A well-drained land produces grazing land early in the rainy season;
- The risk of causing damage to the soil structure by livestock traffic on wet soils is considerably reduced; and
- Good drainage reduces the incidence of diseases which thrive in wet conditions.

Drainage benefits related to soil cultivation: Land cultivation benefits from good drainage as follows:

- Well-drained soils reduce the number of occasions when cultivation is held up waiting for soil to dry up;
- There is less damage to the soil structure; and
- Good drainage allows early and good quality harvest.

7.5 **Types of field drainage systems**

Drainage systems include: 1) Surface drainage; and 2) Sub-surface drainage

Surface drainage: The objective of surface drainage is to increase the removal of surface water either by eliminating low spots where the water tends to accumulate or by constructing drains and ditches to speed up the movement of water out of the field.

Surface drainage systems include:

- Random or irregular drains: This system is suited to areas which have depressions too deep or too large to be filled by land levelling. The surface drains meander from one slow spot to another, collecting water and carrying it to the lateral outlet drain and from there to the main outfall drain. The drains should be of sufficient sizes to be able to drain the impounded water quality.

- Interceptor drain: These are used to prevent runoff from entering the field from outside especially from the upper portion of slope. They are also constructed across the predominant slope of the land. In practice, they are constructed at about 30 m to 45 m intervals. They are triangular shallow but wide drains: 5 – 8 m wide and 0.15 – 0.25 m deep. All areas between the drains should be levelled and all depressions removed. This method is best suited foe sloping wet fields of 4 % slope or less where the natural drainage system is poor due to compaction of the soil.
- Diversion drain: This is suited to flat, poorly-drained soils which have numerous shallow depressions. They are normally about 180 m to 360 m apart and the areas between the drains are to be levelled and all depressions eliminated.
- Field drains: These are a system of parallel ditch drains constructed at fixed interval depending on the soil type to control both surface and subsurface drainage.

Capacity of surface drains: The capacity of surface drains can be estimated using the Manning's equation as:

$$Q = \frac{A}{n} R^{2/3} S^{1/2} \tag{7.1}$$

Where,
Q = capacity of surface drain (m^3/s)
A = area of flow in the drainage channel (m^2)
R = hydraulic radius of drainage channel (m)
S = bed slope of the drainage channel

Appropriate channel dimensions can then be determined to carry the surface drain flow.

Construction of open drains: Good quality construction is required for effective drainage and can best be insured but having the right type of personnel to carry out the work. Site conditions such as soil type, soil stability, soil wetness, size of drains to be constructed etc. affect the type of equipment to be used. Apart from using manual labour, various types of equipment are available including dragline, scraper, backhoe, grader, bulldozer, shovel and plough. To prevent serious erosion and to stabilize the drain bottom and side walls, vegetation should be established on these areas as soon as practicable, soil and climate permitting.

Maintenance of drains: A good maintenance programme is as essential as proper design and construction. Maintenance is a continuing requirement if the drain is expected to continue to perform as designed. Preventive maintenance usually begins immediately following construction and corrective maintenance follows any subsequent failure of drainage system or any part of it. For example, the Manning's roughness coefficient of an unlined drain can rise from a value of 0.02 immediately after construction up to 0.08 or more when infested by weed growth. The entire drainage system should be inspected during or immediately following a heavy runoff event soon after completion of construction and at least annually thereafter. Structures should be checked frequently for obstructions, excessive erosion and any other defects.

Sub-surface drainage: This aims at increasing the rate at which water will drain from the soil, and so lowering the water table and increasing the depth of drier soil above the water table.

Sub-surface drainage systems include: 1) Ditch drains; and 2) Pipe or tube drains.

Ditch drains are deep parallel open drains primarily to control subsurface water but also control surface water. The construction of the ditches take some land out of production. On the other hand, pipe or tube drains do not take land out of production. The parallel pipe drains are pipes buried underground and usually enveloped in filter material such as sand and gravel. Sub-surface drainage theory is based on the Hooghoudt Steady State Drainage Equation.

For ditch drains, the Hooghoudt equation is written as:

$$\boldsymbol{q = \frac{8kDh}{L^2} + \frac{4kh^2}{L^2}} \tag{7.2}$$

Where,

q = drainage coefficient (m/d)
k= average soil hydraulic conductivity (m/d)
h= mid-drain water table height above level of water in drain (m)
D= depth of water in drain (m)
L = drain spacing (m)

The drainage coefficient is expressed as depth of water over a drained area which is to be removed in 24 hours or a day. Conservatively it can be estimated as:

$$\boldsymbol{q = \frac{P}{100}} \tag{7.3}$$

Where,
P = mean annual rainfall (mm). Equation 8.2 applies for mean annual rainfall of more than 1000 mm; else q should be taken as 10 mm/d.

Hooghoudt Steady State Drainage Equation for pipe or tube drains is given as:

$$\boldsymbol{q = \frac{8k_b h d_e}{L^2} + \frac{4k_a h^2}{L^2}} \tag{7.4}$$

Where,

q = drainage coefficient (m/d)
k_a= average soil hydraulic conductivity above the level of pipe drain (m/d)
k_b= average soil hydraulic conductivity below the level of pipe drain (m/d)
h= mid-drain water table height above level of pipe drain (m)
d_e= depth of water in drain (m)
L = drain spacing (m)

and

$$\boldsymbol{d_e = \frac{D}{1+\frac{2.54D}{L} ln\left(\frac{0.258D}{r_o}\right)}} \tag{7.5}$$

Where,

D = depth of the pipe drain to the impermeable layer (m)
r_o= radius of pipe drain (m)

Equation 8.5 is known as the van Beers equation. Equations 8.14 and 8.15 are used iteratively to compute the drain spacing.

Resultant hydraulic conductivity for layered soils: Assume layered soils 1, 2, 3, 4, ……, m of thickness d_1, d_2, d_3, d_4, ……, d_m with corresponding hydraulic conductivities k_1, k_2, k_3, k_4, ……, k_m. Assume further that a pipe drain is installed at the boundary between layers 3 and 4, then the resultant hydraulic conductivity above the level of tube drain (k_a) can be computed as:

$$\boldsymbol{k_a = \frac{k_1 d_1 + k_2 d_2 + k_3 d_3}{d_1 + d_2 + d_3}} \quad (7.6)$$

Generally, this equation can be written as:

$$\boldsymbol{k_a = \frac{\sum_{i=1}^{n} k_i d_i}{\sum_{i=1}^{n} d_i}} \quad (7.7)$$

Where,
n = number of layers above the drain

Resultant hydraulic conductivity below the level of tube drain (k_b) is given as:

$$\boldsymbol{k_b = k_{n+1}} \quad (7.8)$$

Where,
k_{n+1} = hydraulic conductivity of the soil layer immediately below the drain.

Re-computation of *D* as used in the van Beers equation: If the resultant hydraulic conductivity below the drain is equal to the hydraulic conductivity of the soil layer immediately below the drain, then it will require the re-computation of the value *D*, which is the depth of the drain to impermeable layer as used in equation 8.5. The re-computed *D* is given as:

$$\boldsymbol{D = \frac{\sum_{i=n+1}^{m} k_i d_i}{k_b}} \quad (7.9)$$

Where,
m = total number of soil layers in the profile below drain down to impermeable layer.

7.6 **Reclamation of Salt-Affected Lands**

Salt build up in the soil can be harmful to the crop and either drastically reduce yields or completely destroy the crop. Crops are sensitive to salt accumulation in the soil and controlling the salt levels is usually a compromise between accepting some yield reductions and the cost of salt control operations. The salt

problems are 1) salinity; and 2) sodicity. Crops are sensitive to chlorides, sodium and boron. Sodium salts are toxic to crops and tend to reduce the permeability of the soil. Different crops have different levels of tolerance to salinity and sodicity. Forage crops are the most resistant to salinity, followed by field crops, vegetable crops, and fruit crops. Salinity problems are indicated by the electrical conductivity of the soil saturated extract (EC_e). This is measured in mmhos/cm (dS/cm). Sodicity problems are measured by the sodium adsorption ratio (*SAR*). The unit for SAR is $(\text{mmol/l})^{1/2}$.

SAR can be computed as:

$$\boldsymbol{SAR} = \frac{\boldsymbol{Na^{+}}}{[(\boldsymbol{Ca^{2+}} + \boldsymbol{Mg^{2+}})/\boldsymbol{2}]^{\boldsymbol{1/2}}} \tag{7.10}$$

A soil is said is said to be saline or sodic according to the following classification:

Classification	Normal	Saline	Sodic	Saline-sodic
EC_e (mmhos/cm)	< 4	> 4	< 4	> 4
SAR $(\text{mmol/l})^{1/2}$	< 13	< 13	> 13	> 13

Salinity and Sodicity Control: This can be achieved through 1) soil amendment and 2) leaching. Soil amendment involves incorporating large amounts of other chemicals into the soil to neutralize the effect of those causing the salinity and sodicity problems. The costs involved include cost of chemicals and the cost of incorporation in the soil. So, the costs can be high. The other option of leaching is based on the fact that most salts are water soluble and therefore applying large amounts of irrigation water to the soil to leach the salts. The management decisions will involve 1) determining the leaching fractions required; and 2) how long the irrigation must be applied to leach the soil.

Leaching fraction: For salinity control, the leaching fraction (LF) can be computed as:

$$\boldsymbol{LF} = \frac{\boldsymbol{D_{dw}}}{\boldsymbol{D_{iw}}} = \frac{\boldsymbol{EC_{iw}}}{\boldsymbol{EC_{dw}}} \tag{7.11}$$

Where,
D_{dw} = depth of irrigation water (mm)
D_{iw} = depth of drainage water (mm)
EC_{iw} = electrical conductivity of irrigation water (mmhos/cm)
EC_{dw} = electrical conductivity of drainage water (mmhos/cm)

The electrical conductivity of the drainage water is also equivalent to the electrical conductivity of soil saturated extract(EC_e).

Another leaching fraction formula after Rhoades is given as:

$$\boldsymbol{LF} = \frac{\boldsymbol{EC_{iw}}}{\boldsymbol{5EC_e - EC_{iw}}} \tag{7.12}$$

For sodicity control, the leaching fraction formula is:

$$\boldsymbol{SAR_{dw} = SAR_{iw}\left[\frac{Y^{(1-2LF)}}{\sqrt{LF}}\right][1 + (8.4 - pH_c)]} \quad (7.13)$$

Where,
SAR_{dw}= sodium adsorption ratio for drainage water (mmol/l)$^{1/2}$
SAR_{iw}= sodium adsorption ratio for irrigation water (mmol/l)$^{1/2}$
Y = soil mineral weathering coefficient = 0.7

The term $(8.4 - pH_c)$ indicates the tendency of the applied water to precipitate or dissolve $CaCO_3$. When this term is greater than 0, $CaCO_3$ precipitates in the soil when water is applied. When the term is less than 0, $CaCO_3$ dissolves in irrigation water. pH_c is computed as:

$$\boldsymbol{pH_c = (pK_2 - pK_s) - p\left[\left(Ca^{2+} + Mg^{2+}\right)/2\right] + p\left(HCO_3^- + CO_3^{2-}\right)} \quad (7.14)$$

Where,

pK_2 = - $\log_{10}$ (second dissociation constant of H_2CO_3)
pK_s = - $\log_{10}$ (solubility product constant of $CaCO_3$)

The values of $(\boldsymbol{pK_2 - pK_s})$ as a function of total cation concentration are available in standard tables.

Time for leaching: Time for leaching can be computed as:

$$\boldsymbol{t = \frac{ET_c\ x\ II}{(1-LF)i_{av}}} \quad (7.15)$$

Where,
t = time required for leaching (h)
ET_c = crop evapotranspiration (mm/d)
II = irrigation interval (d)
LF = leaching fraction
i_{av}= average infiltration rate (mm/h)

7.7 **Exercises**

1. State whether the following statements are ***True*** or ***False*** concerning the benefits of agricultural land drainage:
   a) Good drainage leads to a limited choice of crops for the land.
   b) A well-drained soil warms up more quickly.
   c) Lower water table results in an increased volume of soil available for plant growth.
   d) The risk of causing damage to the soil structure by livestock walking on wet soil is considerably reduced when the soil is well-drained.
   e) Well-drained land increases the incidence of animal diseases that thrive in wet conditions.
   f) Good drainage reduces the number of occasions when cultivation is held up waiting for the soil to dry up.
   g) Good drainage allows early and good quality harvest.
   h) Land cultivation in poorly-drained soils cause less damage to soil structure.

2. The purpose of *subsurface* drainage is to _____

3. Name the surface and subsurface drainage systems you know.

4. In the construction of open drains, name 4 site conditions that affect the type of equipment to be used.

Problem: A subsurface drainage system is to be installed in an irrigated area to facilitate leaching. Determine the required drain spacing according to the *Houghoudt* Method for steady state conditions for the following fields:

Parameters	**Field 1**	**Field 2**
k_1	0.38 m/d	0.50 m/d
k_2	1.48 m/d	1.00 m/d
D	2.40 m	3.00 m
h	0.50 m	0.60 m
q	7 mm/d	5 mm/d
r_o	0.10 m	0.10 m

Answer the next 2 questions.

5. Determine the drain spacing for *field 1*.

6. Determine the drain spacing for *field 2.*

7. State 2 parameters (and their units) that are used to describe salt-affected lands for the purposes of their classification.
   a)
   b)

8. A soil is said to be *sodic* if it fulfils which condition of soil extract concentration? _____

9. A soil is said to be *normal* if it fulfils which condition of soil extract concentration? _____

Problem: Classify the following soils for the purpose of their reclamation, if necessary, using the data obtained from their analysis:

Soil A

ECe = 5.85 mmhos/cm Ca^{2+} = 1.41 mmol/l

Mg^{2+}= 1.01 mmol/l Na^{2+} = 10.5 mmol/l

Soil B

ECe = 3.80 Ca^{2+} = 1.42

Mg^{2+} = 1.21 Na^{2+} = 22.50

Answer the next 2 questions.

10. Soil A can be classified as _____

11. Soil B can be classified as: _____

12. Calculate the required time of leaching of a salt-affected land for the following data using the *two* different leaching fraction formulae:

Crop	: Tomatoes
EC_d	= 5.5 mmhos/cm
EC_{iw}	= 3.2 mmhos/cm
Soil infiltration rate	= 10 mm/h
Crop evapotranspiration	= 7.5 mm/d
Irrigation interval	= 10 d

13. One harmful effect of soil sodicity is that sodium ions tend to seal up the soil pores thereby reducing the permeability of the soil. (***True*** or ***False***?)

14. Maintenance of surface drains is a continuing requirement if the drain is expected to continue to perform as designed. List the types of maintenance procedures you can use.

15. (a) Explain how you will determine the resultant hydraulic conductivity above and below the drain in a multi-layered soil (each layer having different hydraulic conductivity) for use in the Hooghoudt steady-state drainage equation.

    (b) Below is the data for hydraulic conductivity of multi-layered soil with tube drain of radius 0.1 m installed at 1.0 m below ground level in a farm. The average rooting depth of the crop grown in the farm is 0.70 m. What drain spacing (m) is required to control the water table if drainage coefficient is 5 mm/d?

Soil depth (m)	**Hydraulic conductivity (m/d)**
0-0.4	0.5
0.4-0.7	0.4
0.7-1.0	0.7
1.0-1.3	0.7
1.3-2.0	1.2
2.0	Impermeable layer